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AI and Blockchain Integration in Smart Grids for Autonomous Energy Trading and Risk Optimization

GridMind
Engineering thesis manuscript

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Empirical scope: community battery control and settlement using measured Irish profiles

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Streszczenie

Praca przedstawia prototyp GridMind służący do oceny lokalnego współdzielenia energii, sterowania magazynami energii i rozliczania transakcji w społeczności sześciu gospodarstw domowych. Wykorzystano zmierzone profile zużycia energii i generacji fotowoltaicznej ze zbioru StoreNet w Irlandii. Taryfy, parametry akumulatorów i koszty ich eksploatacji przyjęto jako założenia scenariusza.

Badanie obejmuje trzy odrębne porównania. Pierwsze dotyczy podziału korzyści za pomocą alokacji proporcjonalnej, programowania liniowego i regularyzowanego kryterium Nasha. Drugie porównuje sterowanie bezczynne, regułę autokonsumpcji, sterowanie predykcyjne MPC oraz wyuczone polityki IPPO i MAPPO. Trzecie odtwarza ten sam strumień rozliczeń w lokalnym kontrakcie Solidity i podpisanej bazie SQLite.

Pierwotny koszt MPC wyniósł 159,1604 EUR wobec 159,4083 EUR dla nieaktywnego magazynu. Średnie koszty IPPO i MAPPO wyniosły 173,3106 i 182,8197 EUR. Wykonano 60 dodatkowych przebiegów dotyczących nagrody, rodzaju akcji i budżetu treningu. Wszystkie dziesięć modeli dyskretnych przy 300 aktualizacjach odtworzyło wynik nieaktywnego magazynu. Połączenie prognozy Ridge z MPC obniżyło koszt do 158,8618 EUR, lecz dwa gospodarstwa poniosły stratę względem wariantu bazowego. Oba rejestry rozliczeń uzyskały identyczne salda po 64 transakcjach.

Wyniki czerwcowe pozostają eksploracyjne. Ocena zamrożonych modeli obejmuje dodatkowo 501 czterogodzinnych epizodów z okresu lipiec–grudzień, w 168 dniach z zachowanymi obserwacjami. Koszty wariantu bezczynnego, MPC z prognozą trwałości i MPC z Ridge wynoszą odpowiednio 1947.4997, 1946.6754 i 1939.3432 EUR. Wcześniejszy dostęp do części wyników wyklucza interpretację jako niezależne potwierdzenie. Nie przeprowadzono oceny na polskich danych rynkowych ani sterowania z dziennym ograniczeniem CVaR. Wkładem jest odtwarzalna ocena z jawnymi ograniczeniami wnioskowania.

Słowa kluczowe: społeczność energetyczna, magazyn energii, MPC, uczenie wieloagentowe, sprawiedliwość podziału, blockchain.

Abstract

This thesis evaluates local energy sharing, battery control and transaction accounting for six households using measured Irish StoreNet load and photovoltaic profiles. Tariffs, batteries and throughput costs are scenario assumptions. The work separates allocation fairness, intertemporal control and ledger verification so that each comparison has a defined baseline.

The allocation pilot compares proportional sharing, a welfare linear program and a regularized Nash objective. The battery experiment trains IPPO and MAPPO with five seeds each and compares them with idle sharing, a self-consumption rule and causal MPC. Training uses 404 complete four-hour episodes, validation uses 105 and testing uses 44. All batteries start and finish each episode at the same energy, and policy observations exclude current target and future measurements.

In the original experiment, persistence MPC costs 159.1604 EUR against 159.4083 EUR for idle sharing. Mean IPPO and MAPPO costs are 173.3106 and 182.8197 EUR. Sixty follow-up runs vary action representation, training budget and a fairness penalty. All ten discrete policies at 300 updates reproduce idle cost. A new forecast-to-MPC experiment reduces cost to 158.8618 EUR with Ridge, while two households lose relative to idle. A local Solidity contract and signed SQLite ledger agree over 64 settlements.

The June results remain exploratory. A frozen-model extension evaluates 501 four-hour episodes from July–December across 168 retained days. Idle, persistence MPC and Ridge MPC cost 1947.4997, 1946.6754 and 1939.3432 EUR respectively. Its documented prior access prevents a confirmatory claim. Incomplete coverage, six households and short horizons limit inference. No Polish trading outcome or enforced daily CVaR constraint is reported. The contribution is reproducible evaluation with causal forecast-to-control coupling, controlled interventions and explicit limits.

Keywords: energy community, battery control, multi-agent reinforcement learning, MPC, fairness, smart contracts.

Abbreviations and notation

AC — Alternating current

BESS — Battery energy storage system

CVaR — Conditional value at risk

EVM — Ethereum Virtual Machine

IPPO — Independent proximal policy optimization

LP — Linear program

MAE — Mean absolute error

MAPPO — Multi-agent proximal policy optimization with a centralized critic

MPC — Model predictive control

P2P — Peer-to-peer

PV — Photovoltaic generation

SOC — State of charge

kW and kWh — Power and energy respectively

L and G — Interval consumption and photovoltaic generation

C and U — Cost and household benefit

J — Jain index for a declared benefit quantity

1 Introduction

1.1 Problem and scope

1.1.1 Primary question

The intended application targets a price-taking battery trader in the Polish bidding zone. The completed empirical demonstrator uses six Irish household load and PV profiles to evaluate local sharing, battery control and transaction accounting. These are different economic settings. The demonstrator provides software and methodological evidence; it does not estimate Polish wholesale trading profit. The thesis distinguishes this measured evidence from the proposed market application throughout.

The empirical question is how control cost and household outcomes change with critic information, action representation, training budget, reward and causal forecasts in the measured community case. The intended application asks whether forecast-informed battery trading can improve net value in the Polish market. The latter requires an acquired and audited market dataset and remains untested. A 95 percent savings-retention criterion belongs to the community study and is not transferred to wholesale profit without redefining the comparator.

1.1.2 Research questions and evidence status

Table 1.1 Research questions and evidence status

Question	Evidence and interpretation
Community cost	Original baselines and 60 follow-up training runs; June comparisons and the later frozen-model evaluation in Chapter 6.
Distribution	Allocation pilot and signed-benefit reward comparison; no universal fairness or retention guarantee.
Risk	Fractional-tail episode CVaR diagnostic; no enforced daily constraint.
Settlement	64 fixed transactions with balance parity and rejection checks; local trust assumptions remain.

Source: implemented experiments and retained records. AI-assisted preparation.

Ireland is the geographical source of all measured household experiments in Chapters 5 –7. Poland is the target of the subsequent market design described in Sections 8.10–8.12. The Irish study minimizes community net cost under assumed tariffs; the proposed Polish trader maximizes net arbitrage profit under market prices. No scaling, currency conversion or relabeling makes those objectives empirically interchangeable.

Core work comprises a validated simulator, fair and economic baselines, a causal forecast/control comparison, household-level outcomes and reproducibility. Battery control supplies a meaningful intertemporal decision problem for learned agents. A full blockchain network and feeder power-flow validation are bounded extensions, consistent with the previous roadmap; their absence must remain visible in the final claims.

1.2 Operational research questions

The research questions distinguish implementation correctness, economic performance and distribution. Table 1.2 records the modeling distinctions needed to interpret the experiments. Completed comparisons are reported in Chapters 5–7; unresolved external-validity and risk-constraint questions are discussed in Chapter 8.

Table 1.2 Operational research questions

Earlier formulation	Revised formulation
Community structure makes blockchain necessary	Define who writes, who is trusted and what failure must be resisted. Compare a signed database with a ledger; community membership alone does not prove necessity.
Nash utility can be applied to any profit	Define savings relative to a grid-only outside option. Specify positivity, normalization and regularization before taking logarithms.
Jain and Gini establish fairness by themselves	Declare whose benefits, over which time horizon, and with which normalization. Report losers, zero-gain participants and raw monetary outcomes.
All energy trading uses marginal uniform pricing	Distinguish the day-ahead auction, intraday trading and local energy-sharing settlement. This pilot uses an explicitly specified bilateral midpoint.
A 15-minute model is always required	Match the decision to its market and data. Finer resampling does not invent information; 30-minute measurements cannot become independent 15-minute observations.
Rule-based versus proposed AI is sufficient	Include proportional sharing, an economic LP/MPC baseline and a fairness optimizer, then compare learned policies under equal information.
Recent headline percentages are targets to beat	Do not transfer improvement percentages across different datasets, vehicles, tariffs or baselines. Verify each source and compare methods on the same test set.

Source: research implementation and recorded outputs. AI-assisted preparation.

The original dispatch and mean-forecast functions remain a compatibility scaffold. The research layer adds optimization, a fitted forecast, provenance and accounting tests. The website and GitHub repository are separate from this package.

The pilot also motivates an additional correction: when every household has the same outside tariffs and total matched energy is fixed, changing allocation changes distribution but cannot change total community savings. A study that ignores this identity risks interpreting a numerical artifact as an economic contribution.

1.3 Computational assistance and research provenance

ChatGPT and Codex (OpenAI) assisted the drafting and editing of all chapters and abstracts, the implementation and execution of research code, numerical analysis, tables, figures and defense materials. These are AI-assisted parts of the work throughout, including the additions in Sections 4.7 and 6.12–6.14. Numerical findings reported as executed are tied to retained data, scripts and result files; generated prose is not independent evidence. Appendix B records the scope of assistance and the team contribution record.

PUT Senate Resolution 195/2020–2024 requires explicit disclosure and prior consultation with the supervisor about GenAI use (Poznań University of Technology, 2024). This manuscript does not attest that consultation or independent student verification has occurred. Each student remains responsible for reviewing the analysis, developing their own conclusions and meeting the supervisor’s authorship requirements before submission.

2 Research foundations and market context

2.1 Related research

2.1.1 Forecasting and control

The literature supports separate foundations for forecasting, control, fairness and trust. Their integration should be evaluated through controlled comparisons; combining familiar methods is not, by itself, evidence of a new scientific result.

Lago et al. provide an open electricity-price forecasting benchmark and emphasize defensible comparative evaluation (Lago et al., 2021). GridMind adopts the methodological principle of strong simple baselines, chronological evaluation and reported error metrics. Its current pilot forecasts aggregate net energy one step ahead, so it must not inherit performance claims from a day-ahead price benchmark. CityLearn offers a reference environment for comparing rule-based, model-predictive and reinforcement-learning energy control (Nweye et al., 2024). Its use does not automatically validate an Irish market model.

2.1.2 Learning across multiple households

Yu et al. demonstrate the effectiveness of PPO-based methods in cooperative multi-agent testbeds (Yu et al., 2022). This supports using MAPPO as a candidate comparison, but it is not evidence that MAPPO will outperform optimization in this energy community. A fair test requires the same observations, feasible actions, horizon and training budget. The independent-agent and cooperative-agent variants should differ in the intended coordination mechanism, not in privileged access to future data.

2.1.3 Fairness and risk

Jain, Chiu and Hawe provide a bounded resource-allocation index (Jain, Chiu and Hawe, 1984). In GridMind, applying it to normalized bill savings is a modelling decision that must be defended separately. Rockafellar and Uryasev provide an optimization representation of conditional value at risk (Rockafellar and Uryasev, 2000). GridMind uses this representation to specify a future daily-cost risk constraint and implements the empirical fractional-tail calculation for diagnostics.

2.1.4 Data and trust

StoreNet provides a relevant Irish community measurement source (Trivedi et al., 2024; Khadem et al., 2024). Wüst and Gervais provide a decision framework for assessing whether a blockchain is appropriate (Wüst and Gervais, 2018). Their framework

motivates a trust-model comparison, rather than a blanket assertion that a community needs a blockchain.

2.1.5 Proposed contribution

The contribution is a reproducible evaluation of allocation, battery control and settlement, including controlled reward and action/budget interventions. It establishes descriptive outcomes under the recorded protocols and retains household losses. It does not identify a universal best controller or the single dominant cause of learning failure. Independent evaluation and further matched interventions are needed for those stronger claims.

2.1.6 Evaluation reliability and the research gap

Schulman et al. introduce PPO as an alternating procedure of collecting interaction data and optimizing a surrogate objective over multiple epochs (Schulman et al., 2017). GridMind uses a clipped policy-ratio objective, a value loss and an entropy term. This identifies the algorithm family; the particular observation set, reward scale and stopping rule remain implementation choices requiring evaluation in the energy setting.

Henderson et al. examine reproducibility and variation in deep reinforcement learning, showing why implementation and reporting choices can change apparent rankings (Henderson et al., 2018). This motivates retaining every seed, checkpoint-selection history and unsuccessful run in GridMind. A low final training loss is not treated as evidence of economic value, and a selected test return is not substituted for an account of how the model was chosen.

Agarwal et al. show that few-run reinforcement-learning benchmarks can produce unstable conclusions when reporting point estimates alone (Agarwal et al., 2021). GridMind consequently labels seed summaries as descriptive and separates training variability from repeated observations of the same households. The present sample does not support a population-level superiority claim. A future independent study needs temporal uncertainty analysis and a sampling design appropriate to the intended generalization.

The resulting research gap is specific: whether a controller can improve the distribution and level of benefits in this measured community under a causal information boundary, and whether its settlement layer adds a needed trust property. This thesis supplies reproducible comparisons and controlled interventions for that case. It introduces neither a new PPO algorithm nor a new cryptographic protocol. Its contribution is the

evidence linking modeling choices to economic outcomes and the conditions under which those conclusions stop applying.

2.2 Market boundary and data provenance

European day-ahead market coupling has transitioned to 15-minute market time units, with the first delivery day on 1 October 2025 (ENTSO-E, n.d.). This informs a contemporary wholesale extension. It does not mean that a local study using older household measurements represents the same market regime, or that all European settlement rules are identical.

The European Commission describes updated energy-sharing rights applying from 17 July 2026 and the role of national implementation (European Commission, n.d.). GridMind should distinguish local sharing among members from the community's external supplier or balance-responsible arrangement. A smart contract does not itself create the right to supply electricity, alter metering responsibilities or remove network charges. These are research-boundary statements; operational compliance is outside the pilot.

Table 2.1 Market boundary and data provenance

Data source	Use and boundary
StoreNet Ireland	Primary measured load/PV case. The delivered pilot uses H1-H6. It does not claim that these six households represent all Irish communities.
CER Ireland	Alternative cohort not used in this experiment; access and data suitability require an independent audit.
OPSD households	Secondary German cross-check (Open Power System Data, 2020). Preserve field definitions, units and interpolation flags; its historical period differs from current market rules.
External price series	Add only with verified region, period, currency, interval and publication time. Current assumed retail tariffs are scenario inputs, not observed prices.

Source: research implementation and recorded outputs. AI-assisted preparation.

Three evidence categories

Measured inputs are published consumption and PV series. Scenario assumptions are tariffs, fees, battery parameters and the single-bus representation. Chapters 6 and 7 evaluate control and local settlement. Wholesale bidding remains outside the implemented scope. These categories are kept separate when interpreting the result tables.

For a historical replication, use rules and prices appropriate to the measurement period. For a contemporary policy scenario, state that historical physical profiles are replayed under a later rule set. Do not mix these interpretations. If source timestamps cannot be reconciled to a verified time zone, retain their original labels and avoid external price/weather joins. This is the approach taken in the present pilot.

The data register is designed to retain the original file, source DOI, license, download date, hash, household identifier, native interval, time-zone convention, measured quantity, unit and exclusion reasons. Downloaded files and local derivatives must remain distinguishable. Expanding six profiles into 50 or 100 simulated agents is a scaling exercise, not an increase in independent real-world observations.

3 Physical economic and fairness model

3.1 Energy and accounting model

Let L_{it} and G_{it} denote household consumption and PV generation in kWh during interval t. Positive net energy n_{it} is a deficit. Let b_{it} and s_{it} be the external import and export tariffs in EUR/kWh. Each household's outside option uses the same load, PV and tariff data as the sharing policy.

$$n_{it} = L_{it} - G_{it}; \quad d_{it} = \max(n_{it}, 0); \quad a_{it} = \max(-n_{it}, 0) \tag{3.1}$$

For seller i and buyer j, x_{ijt} is allocated energy in kWh. The current feasible set requires nonnegative allocations, seller allocations no larger than surplus, and buyer receipts no larger than deficit. A household cannot be both buyer and seller within the interval. All households are represented at one unconstrained bus; this checks energy balance, not feeder voltage or thermal feasibility.

$$x_{ijt} \geq 0; \quad \sum_j x_{ijt} \leq a_{it}; \quad \sum_i x_{ijt} \leq d_{jt} \tag{3.2}$$

The pilot uses bilateral midpoint pricing $p_{ijt} = (b_{jt} + s_{it})/2$. A fee f in EUR/kWh is paid outside the community and split equally between buyer and seller. A trade is admissible only when $b_{jt} - s_{it} - f$ is positive. Each side then gains half of the available spread times the energy traded, ensuring individual rationality for this simplified allocation model.

$$u_{i \leftarrow j,\, t} = u_{j \leftarrow i,\, t} = \tfrac{1}{2}(b_{jt} - s_{it} - f)x_{ijt} \tag{3.3}$$

Let C^0 be the total external cost under grid-only exchange. Let C be the external cost after local allocation, including the external sharing fee. Summed household savings W equal $C^0 - C$. Internal payments cancel because every local purchase is another household's receipt. The code verifies this identity for every interval.

$$W_t = \sum_i u_{it} = C_t^0 - C_t = \sum_{i,j} (b_{jt} - s_{it} - f)x_{ijt} \tag{3.4}$$

An invariant that determines the experiment

If b, s and f are common across households, the equation reduces to $W_t = (b - s - f)\, Q_t$, where Q_t is total matched kWh. Therefore, two policies that match the same volume have identical total savings even if their benefit distributions differ. The pilot verifies this numerically. Heterogeneous outside tariffs, intertemporal flexibility, transaction losses or volume restrictions can create a nontrivial efficiency–distribution relationship.

The term welfare in this pilot means bill savings under declared tariffs. It is not a measurement of unobserved private valuations, comfort preferences, social welfare in the full economic sense or willingness to pay.

3.2 Allocation fairness

Define U_i as household i's cumulative savings over the declared evaluation window. Report U_i in EUR before applying any normalization. The pilot also uses $z_i = U_i/k_i$, where k_i is a positive household scale determined from January–May observations: mean available interval load multiplied by 96 and by a fixed 0.30 EUR/kWh reference tariff, with a 0.10 EUR floor. This is a load-based scale, not household income or vulnerability.

$$J(z) = \frac{(\sum_i z_i)^2}{N\sum_i z_i^2} \tag{3.5}$$

$$\mathrm{Gini}(z) = \frac{\sum_i \sum_j |z_i - z_j|}{2N\sum_i z_i} \tag{3.6}$$

For nonnegative nonzero benefits, larger Jain and smaller Gini indicate greater equality in the chosen quantity. When all benefits are zero, the ratios are undefined; the implementation reports no value, not perfect fairness. Material negative savings are retained in the monetary results, with a negative-benefit rate, and these fairness summaries are marked outside their chosen interpretation domain. They are never hidden by clipping or by adding an arbitrary constant.

Regularized Nash allocation

At each observed interval, the welfare LP first obtains W^*_t, the maximum feasible savings. The fair allocator then maximizes the mean log of normalized cumulative daily savings, including the proposed current allocation. The constraint preserves at least a fraction $1 - \delta$ of W^*_t. In the pilot δ is 0, 0.05 or 0.10; ε is 0.01 on the dimensionless normalized-benefit scale.

$$\text{maximize}_x \quad \frac{1}{N}\sum_i \log\left(\varepsilon + \frac{U_{i,\,\text{previous}} + u_{it}(x)}{k_i}\right) \tag{3.7}$$

$$\text{subject to} \quad x \in \text{feasible allocations}, \quad W_t(x) \geq (1-\delta)W_t^* \tag{3.8}$$

This is regularized Nash allocation, not the unmodified Nash bargaining solution. The daily accumulator resets at midnight in the source clock. The algorithm is myopic: it sees current observed surplus and deficit, but does not optimize future intervals. SLSQP

convergence and primal feasibility are checked; solver failure triggers a labelled feasible LP fallback. The log objective is concave over the feasible set, but the implementation does not independently certify numerical global optimality.

Daily and pooled fairness answer different questions. An average of daily Jain values is generally different from Jain computed on two-week totals. Improving log utility need not improve every Jain/Gini statistic. The final study must fix a primary fairness measure before testing, retain both raw and normalized household outcomes, and report the full trade-off rather than select whichever metric makes a method appear best.

3.3 Battery dynamics and risk formulation

3.3.1 Battery dynamics

A genuine sequential control problem requires a state affected by actions. Battery storage supplies such a state. Without storage, flexible demand or strategic temporal commitments, repeatedly deciding whether to accept an immediately beneficial exchange can be a trivial learning task. Battery dynamics are implemented in Section 6.1. The risk-constrained objective remains proposed; it was not part of the allocation pilot.

Let E_{it} be stored energy in kWh, c_{it} and v_{it} be charge/discharge power in kW, ηc and ηd the efficiencies and Δt the interval duration in hours. Bounds on energy and power are hard constraints. A binary operating mode or a rigorously equivalent formulation prevents simultaneous charging and discharging, including when prices are negative.

$$E_{i,t+1} = E_{it} + \eta_c c_{it} \Delta t - \frac{v_{it} \Delta t}{\eta_d} \quad (3.9)$$

$$E_{\min} \le E_{it} \le E_{\max}; \quad 0 \le c_{it} \le P_c; \quad 0 \le v_{it} \le P_d \quad (3.10)$$

All battery methods start and end each episode at the same stored energy. This prevents a policy from obtaining an apparent saving by depleting unpriced initial inventory. Degradation follows a declared AC-throughput convention. Section 6.8 recalculates costs for fixed dispatch under alternative charges; it does not represent retraining. Grid-only allocation and idle-battery sharing remain distinct outside options.

3.3.2 Risk model

The implemented risk variable is episode cost, including the stated tariff, fee and throughput terms. A future daily formulation would aggregate complete daily external costs and include imbalance charges only if actually modeled. Incremental losses relative to another controller are a different random variable and cannot be interpreted

as the same risk measure. The current four-hour diagnostic does not enforce a daily cost limit.

$$\mathrm{CVaR}_{\beta}(C) = \min_{\tau}\left\{ \tau + \frac{\mathrm{E}[\max(C-\tau,0)]}{1-\beta} \right\} \quad (3.11)$$

A future constrained model could limit CVaR at a declared budget B while optimizing expected saving and distribution. Such a model would require causal scenarios and consistent decisions before uncertainty is observed. The thesis retains the CVaR definition for diagnostics but does not implement this optimization constraint. An oracle with future measurements would be a separate comparator.

The package calculates empirical CVaR with fractional mass at the sample quantile. In the 14-day pilot, a 95 percent tail contains 0.7 day-equivalents and the estimate equals the observed maximum. It cannot validate risk optimization. An initial target of 180 held-out days supplies only nine tail-day equivalents at 95 percent and still requires uncertainty analysis. A reward penalty is not a guarantee that a financial risk constraint is satisfied.

3.4 Interpreting the economics of a battery cycle

A simple analytical cycle helps explain why a controller can move energy correctly yet increase cost. Consider an illustrative battery that absorbs 1 kWh from the AC side. At charge efficiency 0.95, its stored energy increases by 0.95 kWh. Discharging that added inventory through an efficiency of 0.95 delivers 0.9025 kWh back to the AC side. The difference, 0.0975 kWh, is conversion loss. This calculation follows directly from the scenario efficiencies; it is not an additional measured StoreNet experiment and it excludes household demand, photovoltaic generation and local sharing.

Under the throughput convention used in the code, that cycle incurs a charge on 1 plus 0.9025 kWh. At 0.01 EUR/kWh the explicit charge is therefore 0.019025 EUR. If the initial energy is purchased at 0.20 EUR/kWh and the discharged energy avoids an import price of 0.30 EUR/kWh, the avoided purchase is worth 0.27075 EUR. Subtracting the 0.20 EUR acquisition cost and the 0.019025 EUR throughput charge leaves 0.051725 EUR. This arithmetic describes one idealized full-inventory cycle under the stated prices, not the realized benefit of any particular household.

At a constant import price of 0.30 EUR/kWh, the same illustrative cycle avoids only 0.27075 EUR after purchasing 0.30 EUR of energy and paying the same throughput charge. Its net result is −0.048275 EUR. A controller that cycles unnecessarily can therefore produce a physically feasible loss even when it returns to its original stored

energy. The terminal equality removes the value of a free initial inventory from the comparison, but it cannot make an uneconomic cycle profitable. This distinction explains why energy balance and terminal-state tests are necessary but insufficient performance tests.

The cycle also provides a conditional price threshold. When the charging import price is 0.20 EUR/kWh, the later avoided import price must exceed approximately 0.242687 EUR/kWh for this idealized cycle to cover its acquisition and throughput costs. The threshold is obtained by dividing 0.219025 EUR by 0.9025 kWh. It assumes that all discharged energy displaces an import at the later price and that there are no other fees or constraints. If discharge is exported, offsets a different tariff, or changes local sharing, the relevant marginal value changes and this threshold no longer describes the complete decision.

Energy from photovoltaic surplus is not automatically free. In the declared scenario, surplus could otherwise be exported at 0.08 EUR/kWh or allocated locally when another household has a deficit. Charging a battery can therefore forgo an alternative benefit. Conversely, discharging at one house can reduce the volume that another house could have sold locally. The correct economic comparator evaluates the whole exchange pattern, including the outside option and fees, rather than counting only the reduction in one household's grid import. The simulator recalculates settlement after the battery action for this reason.

The meaning of the self-consumption baseline follows from this timing. Its request is the negative of the previous net-energy observation, clipped to the permitted exchange range. It charges after observing a surplus and discharges after observing a deficit. It does not see the current interval's realized load before deciding. A rapidly changing net-energy profile can therefore make its request poorly aligned with the next outcome. Its recorded losses should not be generalized to every possible self-consumption controller, especially one operating with a different measurement and actuation convention.

The battery's usable stored-energy interval is 0.2–1.8 kWh, a width of 1.6 kWh. This is distinct from the 2 kWh nominal capacity and from the maximum 0.5 kWh AC request per interval. Charge and discharge efficiencies mean that a 0.5 kWh AC charge adds 0.475 kWh internally, whereas a 0.5 kWh AC discharge removes approximately 0.526316 kWh internally. Confusing AC exchange with internal stored energy would distort both the bounds and the cost calculation. The code converts between these

quantities before updating the state and accounts for the realized AC exchange in settlement.

These calculations clarify which parameters an implementation team would need to obtain from a real installation. Nominal capacity, usable limits, inverter power, conversion efficiency and a degradation convention influence different parts of the decision. A single statement that a system has a 2 kWh battery does not determine its dispatch feasibility or profitability. The present fixed parameters support a reproducible scenario, while the worked examples make their economic consequences inspectable. They should be replaced by verified installation-specific values before any investment or operating recommendation is drawn.

4 Research design and implementation

4.1 System architecture

4.1.1 Agent design and broader study specification

The architecture separates observation, forecasting, physical control, local allocation and settlement. Input and output timestamps define the information available to each component, so a change in forecast, controller or market assumption can be evaluated separately.

Table 4.1 System architecture

Component	Responsibility	Evidence status
Data layer	Convert units, preserve missingness and provenance	Implemented
Forecaster	Aggregate pilot and household MPC forecasts	Chapters 5 and 6.11
Physical controller	Battery actions under hard constraints	Implemented in Chapter 6
Local allocator	Compute feasible exchange and bill savings	Proportional, LP and Nash tested
Learning agents	IPPO/MAPPO policy comparison	Trained in Chapter 6
Settlement layer	Independent verification and payment workflow	Local EVM tested in Chapter 7

Source: research implementation and recorded outputs. AI-assisted preparation.

The original actor uses lagged net energy, stored energy, clock variables, known tariff and household identity. Actions are battery charge/discharge requests. A feasibility layer enforces power, stored-energy and terminal-reachability bounds. Section 6.10 adds own cumulative realized savings and their community mean to both reward arms. No actor reads the current target interval or future meter measurements.

The implemented comparison uses IPPO with local critics and MAPPO with a centralized critic. Both share actors and the same community reward, so IPPO is not a selfish-reward baseline. Data, action representation and evaluation episodes are fixed for the critic comparison. All five seeds are retained; five runs on the same dates do not represent five independent communities or five independent days.

The follow-up reward study is implemented in Section 6.10. It compares the economic team reward with the same reward plus a change in a negative-variance potential of

cumulative signed savings. Both arms use the same fourteen-component observation, architecture, action set, budget and economic checkpoint-selection rule. This isolates the training reward within that design. It does not optimize a raw logarithm of potentially negative savings or guarantee participation.

4.1.2 Role of language models

Language models assisted research preparation as disclosed in Section 1.3. They do not choose runtime battery actions, clear the local market or validate constraints. Explicit equations and numerical checks perform those tasks; interface labels do not constitute trained agents.

4.2 Prospective protocol and implemented scope

The original pilot and the expanded temporal evaluation are exploratory. Independent confirmation requires a separately documented access history and a fixed dataset rule, main comparator, outcome, seed list, tuning budget and stopping rule before results are inspected. The temporal extension in Section 4.7 freezes models and mechanics but does not claim retrospective test independence.

Table 4.2 Prospective protocol and implemented scope

Experiment	Keep fixed	Change
Allocation baseline	Measured current surplus, tariffs, fees	Proportional, economic LP, guarded Nash
Forecast comparison	Split, targets and available history	Persistence, seasonal naive, Ridge, one justified neural model
Control comparison	Forecasts, assets, tariffs, constraints	Rule-based, deterministic MPC, scenario MPC, learned policy
Fairness ablation	Controller and training budget	Fairness weight or welfare tolerance
Robustness	Frozen selected configuration	PV share, storage capacity, costs and forecast disturbance
Settlement ablation	Identical validated trade stream	Signed database versus permissioned ledger

Source: research implementation and recorded outputs. AI-assisted preparation.

A future confirmatory study would use chronological training and validation followed by an independent evaluation period covering several seasons. Transformations would be fitted on training observations alone. Generalization to new households would

require a separate household holdout; temporal extension of the same six profiles does not establish that property.

The complete-case mask is common to every controller. Coverage and exclusion reasons are reported for each candidate block in the extension. This ensures matched comparisons on retained observations but does not correct selection bias. Savings cannot be extrapolated to complete monthly or annual bills without a defensible missing-data model.

The added uncertainty analysis aggregates household costs within source day and uses paired differences with circular moving-block resampling. Block-length sensitivity and absolute effects accompany the point estimates. Model-seed variation is reported separately. These conditional exploratory intervals neither establish equivalence nor constitute a prespecified noninferiority test.

The delivered evidence retains source and processed-data hashes, split boundaries, code, configurations, seeds, model parameters and per-household outcomes. The later-period execution lock additionally records model and code hashes before recomputation. Original unsuccessful runs remain available.

4.3 Pilot data and methods

The pilot downloaded public StoreNet one-minute power CSVs H1_W through H6_W and retained the original files' SHA-256 hashes. It uses the first six numbered households, rather than choosing a cohort based on favorable allocation outcomes. The published dataset contains measurements from an Irish energy community (Trivedi et al., 2024; Khadem et al., 2024); this specific subset and its coverage were inspected directly for the present run.

Table 4.3 Pilot data and methods

Pilot element	Recorded setting
Evaluation window	1–14 June 2020 in the source timestamp clock
Resolution and units	15 one-minute W observations summed and divided by 60,000 to obtain kWh
Common coverage	1,189 / 1,344 intervals retained, or 88.47 percent
Quality exclusions	Missing/non-finite/negative load or PV, incomplete 15-sample intervals; no interpolation

Pilot element	Recorded setting
Physical counterfactual	Measured consumption and PV replayed; historical battery dispatch excluded
Homogeneous tariffs	Import 0.30, export 0.08, fee 0.02 EUR/kWh
Heterogeneous tariffs	H1–H6 import 0.22, 0.26, 0.30, 0.34, 0.38, 0.42; export 0.08; fee 0.02 EUR/kWh
Policies	Proportional, welfare LP, regularized Nash with 0, 5 and 10 percent welfare tolerance

Source: research implementation and recorded outputs. AI-assisted preparation.

Tariffs are analyst-designed assumptions, not observed historical Irish retail offers. The allocation uses realized current surplus and deficit, making it an ex-post sharing baseline. It does not issue a forecast-based day-ahead bid. Its costs refer only to retained intervals. The original timestamp labels have no explicit offset in the downloaded files; no UTC conversion or external weather/price join was assumed.

Five of the six downloaded series contain nonzero PV production; H6 has no observed PV production in its source file. The cohort therefore does not represent an arbitrary balanced sample of PV and non-PV homes. Measured consumption is replayed as exogenous demand; removing historical battery dispatch creates a counterfactual, not a reconstruction of the original control policy.

The implementation checks seller and buyer energy capacities, total net energy balance, individual benefits and the cancellation of internal payments. No Nash-to-LP fallback was triggered in the completed pilot. Successful numerical allocation is evidence about this model’s constraints; it is not evidence of feeder feasibility, lawful settlement or actual electricity delivery.

Sections 4.3, 6.1 and 7.1 specify the original experiments; Sections 6.9–6.11 report reward, action-budget and forecast interventions. Sections 4.7 and 6.12–6.14 add temporal uncertainty and the July–December extension. Daily CVaR control, actual wholesale bidding and an independent external cohort remain unimplemented.

4.4 Decision timing and information available to the controller

The implemented battery experiment separates the time of a decision from the time at which its outcome is measured. At the start of an interval the controller knows the previous measured household net energy, the present battery energy, the clock and the assumed tariff. It then requests charging or discharging. The environment applies the

physical projection and combines the realized battery exchange with the measured net energy of the current interval. Current demand and photovoltaic generation become the lagged observation for the following decision. This order is essential: exposing the current target before the action would turn a forecast-dependent controller into a controller with advance knowledge.

The original actor observation has twelve components. Stored energy is divided by two and lagged net energy is also divided by two. Four shared components encode the sine and cosine of clock time, the fraction of the episode remaining and the current import tariff. Six one-hot components identify the household. Household identity permits one shared network to represent different behavior for different members; it does not provide the actor with their current measurements. The observation does not include household income, a fairness entitlement, a wholesale price forecast, an order book or a language-model response. These exclusions bound the meaning of autonomous trading in the implemented prototype.

The actor outputs five categorical action scores corresponding to requested AC exchanges of −0.50, −0.25, 0, 0.25 and 0.50 kWh. A negative request discharges the battery and a positive request charges it. These quantities are energy over a fifteen-minute interval, not instantaneous power ratings. The largest request corresponds to 2 kW sustained over that interval. Evaluation chooses the largest action score deterministically, whereas training samples from the categorical distribution. Consequently, the evaluation represents a reproducible deployed action rule and not an average over fresh action-sampling randomness.

An action is a request rather than a guarantee of delivered energy. The projection enforces the battery bounds and preserves the ability to return to the terminal energy. A requested discharge can be reduced when the battery is low or when the remaining intervals cannot restore the required energy. Different categorical requests may therefore map to the same realized exchange. This limits the effective action diversity near a bound and near the end of an episode. A controller can receive an unfavorable economic outcome while remaining fully compliant with these constraints; feasibility and decision quality must therefore be reported separately.

The MPC comparator uses the lagged net energies of all six households and their battery states. It extends the latest observed net energy as a persistence forecast through the remaining episode, uses the known scenario tariff schedule, solves the linear program, and applies only its first action. At the next interval it solves again using the new

observation. The future clock and deterministic tariff are known scenario inputs, while future measured demand and generation are not supplied to the optimizer. This is causal prediction with a simple forecast, rather than a perfect-foresight benchmark.

The original IPPO/MAPPO comparison fixes actor inputs, reward, actions and budget while changing critic information. MPC versus RL compares complete controller designs, including community versus local information and continuous versus discrete requests. Section 6.9 separately varies learned action representation and budget. Even continuous learned policies retain a different information boundary and decision procedure from MPC, so their gap is not a pure comparison of optimizers.

4.5 Training selection and the meaning of replication

Each learning update samples twelve complete episodes from the training pool with replacement and executes sixteen joint decisions in each episode. Sixty updates therefore produce 11,520 joint decision steps and 69,120 household transitions per seed. Sampling with replacement means these are training exposures, not distinct measured periods. The same historical episode can be encountered repeatedly. The four optimization epochs reuse the collected batch; multiplying the environment-transition count by four would incorrectly describe gradient reuse as new environmental evidence.

The original algorithms receive the same per-step mean community saving relative to idle sharing. This is a common economic signal, not an individual bill reward. The follow-up fairness arm adds a signed-benefit dispersion potential, while its matched control retains the economic signal. Individual outcomes are recorded in both cases. A common objective, parameter sharing or a penalty does not impose a hard lower bound on any household's benefit.

The original IPPO value network receives a local twelve-component observation for each household. The MAPPO value network receives the concatenated seventy-two-component community observation and produces a community value estimate that is expanded across households. Both actors remain local at evaluation. This is the implemented meaning of centralized training and decentralized action selection. The comparison should not be described as six independently parameterized actor networks, and the centralized critic should not be confused with a centralized actor that chooses all actions jointly.

The optimizer uses four full-batch epochs per update. The recorded implementation clips the probability ratio to the interval 0.8–1.2, weights squared value loss by 0.5,

subtracts an entropy contribution with coefficient 0.01, and clips the gradient norm to 0.5. The return calculation uses an undiscounted episode objective and generalized-advantage weighting of 0.95. These are explicit implementation choices, not a claim that they are optimal for this dataset. The delivered logs preserve the settings and selected models so that a later change can be treated as a new experiment rather than silently replacing the original result.

Validation takes place after updates 10, 20, 30, 40, 50 and 60. Each candidate is evaluated deterministically on the May episodes, and the checkpoint with the largest mean community saving is retained. The implementation updates the retained model only for a strictly larger score, so an equal score keeps the earlier selected checkpoint. The test results describe this validation-selected policy, not necessarily the final training update. Reporting only the last loss value would therefore be insufficient to reproduce the evaluated model.

Five seeds quantify variability of this training-and-selection procedure on one fixed dataset. They are not five communities, five years or five independent replications of Irish household behavior. Similarly, the 264 household-episode records per controller are nested within six repeatedly observed households and forty-four common periods. The 3,432 rows across thirteen controller-seed combinations do not represent 3,432 independent households. The present analysis uses descriptive summaries and preserves the individual runs rather than claiming population-level certainty from the row count.

June 1–14 was used in both the allocation pilot and the battery extension. This reuse compromises independent confirmation; it does not by itself prove training-data leakage. The recorded checkpoint rule uses May validation, but a previously inspected June period cannot become untouched retrospectively. A new test requires an access history, fixed dates, eligibility rules, outcomes and selection rules before results are opened. No such fresh confirmatory result is included here.

4.6 Evidence integrity and failure handling in the research workflow

The research workflow has several boundaries at which errors could change the interpretation of a result. Data ingestion establishes what was measured and in which units. Episode construction decides which periods enter the experiment. Control execution determines which information is available at a decision. Accounting translates realized exchanges into signed costs and benefits. Reporting then aggregates those records into a small number of comparisons. A successful test at one boundary does not

validate the others. The package therefore retains intermediate records rather than only the final presentation tables.

The processed input file and its source manifest provide an identifiable starting point. The hash stored with the control results ties the delivered experiment to that specific processed file. A matching hash establishes byte identity; it does not independently prove the correctness of the source's measurements or their interpretation. The source timestamp labels have no externally verified time-zone offset in this experiment. Since the tariffs are scenario functions of those retained clock labels, the study avoids joining them to an external market-price series whose time convention might be different.

Episode construction requires sixteen complete rows in each aligned four-hour block plus a complete preceding interval. Values are not imputed. The later-period eligibility file records every candidate block, including separate complete-block and complete-lag flags. This identifies computational exclusion reasons, while the physical cause of missing meter observations remains unknown.

The physical projection constrains the realized action even when the requested action would violate the battery envelope. This is a safety mechanism within the model, not a substitute for diagnosing the actor. A policy that repeatedly requests impossible actions may appear feasible because the projection corrects it. Recording both requested and realized exchanges would reveal that dependence. The delivered household-episode file contains costs, benefits and throughput; it does not contain every requested action. The thesis therefore reports demonstrated feasibility while leaving request-level behavioral diagnosis as an explicit follow-up.

Numerical assertions make several implementation failures visible. Settlement checks an energy residual, the episode end checks restoration to the target state, and the MPC implementation rejects an unsuccessful linear-program solution. It also checks whether simultaneous charging and discharging appear in the computed plan above its numerical tolerance. These checks are tailored to the retained scenario. A future extension with negative prices or different incentives would need its optimization formulation and cycling assumptions reviewed rather than relying on the fact that the current positive-price runs passed.

The solver behavior is intentionally explicit. An unsuccessful MPC optimization raises an error instead of silently reporting a cost from an undefined action. The allocation pilot has a separately documented feasible fallback for its own optimizer. These are different components with different failure semantics. A field controller would need a

defined operational fallback when an optimization times out, receives invalid input or cannot communicate with an inverter. The research script does not claim to provide that complete operational policy merely because an idle comparator exists in the evaluation.

The controller totals can be reconstructed from the household-episode CSV without loading neural networks. The analysis groups by method and seed, sums the six household outcomes over forty-four episodes and compares the resulting cost and saving with the retained seed-total file. This is a useful independent arithmetic check because it crosses an aggregation boundary. It cannot detect an error shared by all underlying interval records, so it complements rather than replaces execution tests. The new descriptive analysis script exposes its grouping and tolerances so that the reported counts can be checked separately.

Checksums should be updated only after the intended artifacts have been assembled. The manifest excludes itself from its own list, avoiding a circular definition. Each listed file is identified by a relative path and a SHA-256 digest. A downstream reviewer can compare the delivered bytes against those values before running the code. This supports consistency and traceability of this package; it does not create an independent certification authority, and a manifest distributed by the same author is not proof that the original research process was free from error.

The delivered tests and verification scripts have distinct scopes. The research tests exercise selected allocation, forecasting, battery and learning behavior. The original ledger experiment executes rejection scenarios and compares two implementations. The defense record checker validates retained hashes and totals. The new saved-ledger checker verifies actual retained signatures and reconstructs balances without modifying the database. Keeping these scopes separate prevents a read-only verification pass from being described as fresh model training, a new EVM deployment or a complete security audit.

A practical review can proceed from inexpensive checks to more expensive reproduction. First inspect the protocol and manifest, then recompute descriptive summaries and verify the stored ledger. Next evaluate the saved policies with the declared dependencies. Repeat training only when the purpose is to assess the training process or the effect of a changed setting. This ordering makes discrepancies easier to localize: a mismatch in a saved total should be resolved before it is obscured by a newly generated experiment. The principle is traceable evidence, not maximizing the number of commands run.

4.7 Frozen model temporal evaluation

The extension evaluates 1 July through 31 December 2020 with the same six StoreNet homes, source clock, battery parameters, tariffs and four-hour resets. The six raw file hashes match the original source manifest, and the regenerated prefix through 14 June is numerically identical to the retained input. Ridge is fitted only on January–April; its coefficients and scaling parameters match the saved forecast model within 1×10^{-12}. All sixty ablation checkpoints are loaded without retraining or reselection.

The evidence-access history limits the status of this comparison. A temporary run on 25 September inspected eligibility and evaluated frozen policies and at least two baselines, but its full artifacts were not retained after workspace maintenance. The present run recomputed all outcomes on 28 September. The reconstructed protocol explicitly records that prior access. It preserves the same July–December period and comparisons; it is neither an untouched holdout nor an externally preregistered test. Reported results come from the retained recomputation.

The primary contrast is the mean of the five discrete 300-update IPPO policies minus persistence MPC in community cost. This family was chosen using June as development evidence. Other learned families and forecast substitutions are secondary descriptive contrasts. Six household costs are summed within each retained source day; seeds are averaged before calculating daily differences, so neither households nor seeds inflate the temporal sample size.

For an ordered vector of N daily contrasts, each bootstrap draw samples circular blocks of b consecutive eligible-day positions with replacement and truncates the concatenation to N positions. The 2.5th and 97.5th percentiles of 10,000 resampled means form the reported interval. The recorded primary block length is seven eligible days, with one, two, three and fourteen as sensitivities; the random seed is 25092026. Missing calendar days mean eligible-day blocks are not necessarily calendar-week blocks. The calculation is conditional on these households, measured eligibility and frozen models, and does not correct exploratory model selection or multiplicity.

5 Allocation and forecasting results

5.1 Allocation outcomes

Table values below are computed from the saved trade and household/day logs. Savings are the community total relative to grid-only exchange over retained observations. Daily Jain is the average of daily indices on normalized savings. Pooled Jain is computed after summing each household’s savings across the entire evaluation window.

Table 5.1 Allocation outcomes

Tariffs	Method	Savings EUR	Daily Jain	Pooled Jain
Common	Proportional	18.6317	0.8797	0.9689
Common	Welfare LP	18.6317	0.8227	0.9402
Common	Nash δ 0%	18.6317	0.8413	0.9232
Common	Nash δ 5%	18.6317	0.8413	0.9232
Common	Nash δ 10%	18.6317	0.8413	0.9232
Different	Proportional	24.6755	0.8642	0.9472
Different	Welfare LP	26.0954	0.7741	0.8797
Different	Nash δ 0%	26.0954	0.8005	0.8844
Different	Nash δ 5%	25.8716	0.8187	0.8855
Different	Nash δ 10%	25.7561	0.8089	0.8850

Source: research implementation and recorded outputs. AI-assisted preparation.

What the pilot establishes

Under common tariffs, every tested method generated 18.6317 EUR of savings. This agrees with the accounting identity: all methods matched the same economically beneficial volume. Differences in fairness do not create additional total savings in this setting.

Under different tariffs, the welfare LP generated 26.0954 EUR; the guarded Nash method at $\delta = 0.05$ generated 25.8716 EUR, retaining about 99.14 percent of LP savings. Its mean daily normalized Jain increased from 0.7741 to 0.8187, an absolute increase of 0.0446. This is a result of the stated numerical allocation model and assumptions, not an AI trading result.

Proportional allocation generated 24.6755 EUR and a higher mean daily normalized Jain of 0.8642. It also exceeded the guarded Nash method on pooled Jain. Thus the proposed fairness optimizer does not dominate this simple rule. Its benefit is a different economic–distribution compromise. Increasing δ did not monotonically improve every fairness statistic, consistent with the mismatch between a myopic log objective and alternative evaluation metrics.

These observations make proportional allocation and the welfare LP indispensable final-study baselines. They also motivate comparing myopic and full-horizon fairness objectives before attributing an improvement to reinforcement learning.

5.2 Interpretation and forecasting

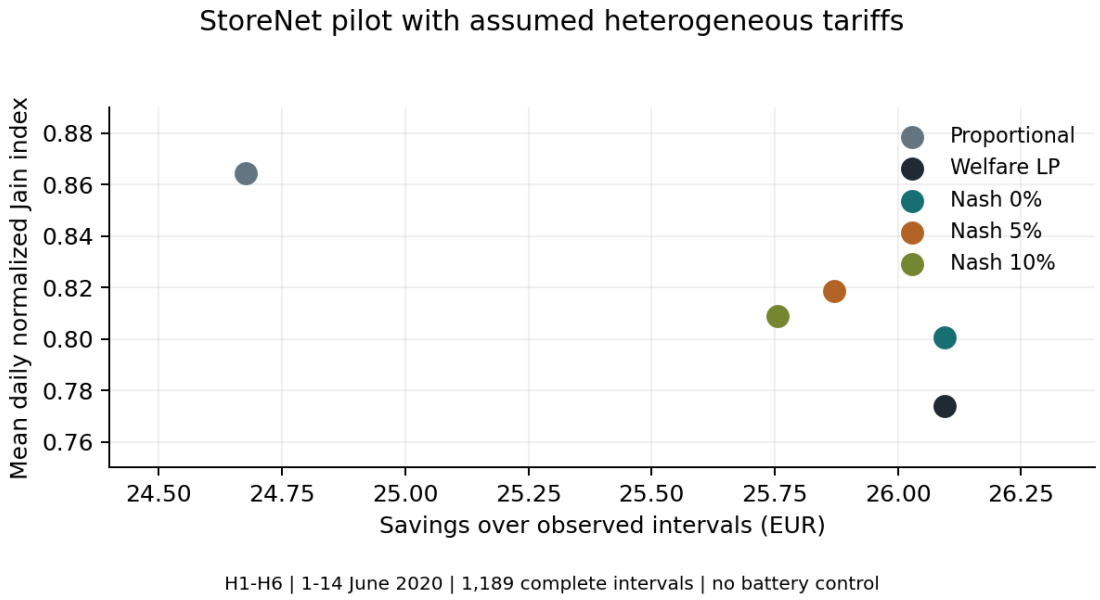


Figure 5.1 Economic savings and mean daily normalized Jain under assumed heterogeneous tariffs. Each point is an evaluated allocation method; the plot is not a learned-policy frontier. Source: research experiment outputs.

The guarded Nash point at a 5 percent interval tolerance sacrifices approximately 0.2238 EUR over the retained two-week observations relative to the LP. Its exploratory paired daily Jain difference has a seven-day moving-block interval of approximately 0.0281 to 0.0611. Only 14 days were available, so this interval is descriptive; it does not validate seasonal generalization or a population effect. Numerical differences near 10^{-15} in the common-tariff savings comparison are floating-point noise.

Separate learned forecasting experiment

The target is aggregate household load minus PV energy in the next 15-minute interval. Features use lags 1, 4, 96 and 672 plus known daily/weekly calendar terms. Training uses January–April 2020; May selects Ridge regularization from 0.1, 1, 10 and 100. The

selected model is refitted on January–May and tested on 1–14 June. All compared methods use the same 883 complete target/lag observations.

Table 5.2 Interpretation and forecasting

Method	MAE kWh	RMSE kWh
Persistence	0.2718	0.3907
Previous day	0.5467	0.7339
Ridge regression	0.2635	0.3691

Source: research implementation and recorded outputs. AI-assisted preparation.

Ridge reduced MAE by approximately 3.04 percent relative to persistence. This small measured improvement is not a formal significance finding. It is a one-step demand/PV-net prediction, not a day-ahead price forecast. Forecasts are not coupled to the allocation pilot, so the experiment establishes no savings attributable to prediction quality.

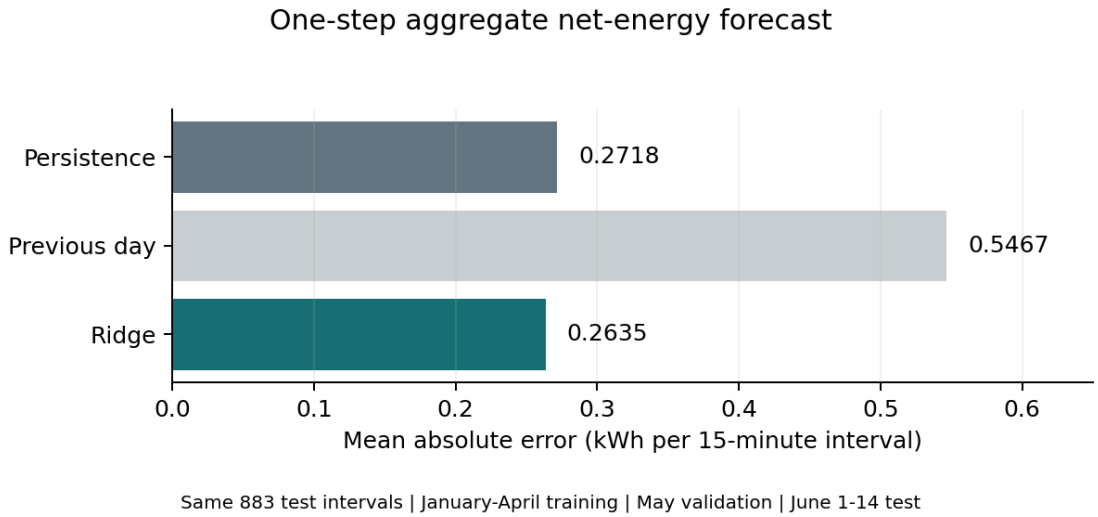


Figure 5.2 Forecast errors from the saved held-out predictions. Lower error indicates better one-step prediction on the selected complete-case test observations. Source: research experiment outputs.

5.3 Verification and limitations

5.3.1 Verification completed

Fourteen research tests passed. They check energy and cash conservation, the homogeneous-tariff invariant, the welfare-retention floor, nonprofitable and negative-price cases, explicit zero/loss handling in fairness metrics, fractional-tail CVaR, paired identical outcomes, invalid inputs, permutation invariance, W-to-kWh conversion, incomplete-interval rejection and prevention of future observations entering current

forecasting features. The assertions cover the new research layer; they do not certify the separate website or the legacy API.

5.3.2 Limits of the empirical evidence

The pilot covers six households, fourteen source-clock days and 88.47 percent of possible allocation intervals. Complete-case selection can change the apparent daily demand or PV mix. The tariffs are assumed, five cohort members have observed PV, and the underlying measurements come from 2020. Outcomes therefore cannot be generalized to today's Irish bills, other countries or a year of operation. The study reports no real market transactions.

The fairness solver is myopic and its daily reset is an explicit design choice. The source-clock offset is not independently verified. Network limits, battery degradation, comfort preferences, private valuations, measurement manipulation and actual settlement fees are not estimated by the current experiment. CVaR is a diagnostic rather than a risk-optimization result. Battery control, trained MARL and the ledger were outside this first pilot; Chapters 6 and 7 report their subsequent evaluation.

5.4 Relating the allocation pilot to the battery extension

The allocation pilot and the battery extension answer related but different questions. The pilot holds measured consumption and generation fixed and changes how local surplus is matched to deficits. The battery extension changes the net exchanges through storage actions while using its own fixed sharing rule. In the first case, the allocation rule redistributes or changes the value of contemporaneous trade. In the second, the controller changes when energy is absorbed or supplied and incurs conversion and throughput costs. Their reported savings cannot be placed in one ranking without aligning these interventions and baselines.

The distinction is especially important for the meaning of total welfare. With common import and export tariffs and the same fee, total sharing saving is the available spread multiplied by matched energy. If two allocation methods match the same volume under those assumptions, changing the pairing cannot create extra community saving. The benefit distribution may still change, which makes fairness analysis meaningful even when the economic total is invariant. A numerical difference in that invariant case would require investigation of unmatched energy, numerical tolerance or inconsistent accounting before it was interpreted as an algorithmic advantage.

Heterogeneous outside tariffs change that situation because the value of a matched unit can depend on the households involved. The pilot's welfare linear program can then favor pairings with larger spreads, while a regularized Nash objective can trade some of that aggregate value for a different benefit distribution. The recorded heterogeneous result illustrates this trade-off: the five-percent Nash configuration retains approximately 99.14 percent of the welfare LP's total saving. The five-percent value is the permitted loss bound in the optimization, not a claim that the realized result must lose exactly five percent.

The fairness comparison also depends on the metric and its aggregation. The reported average daily normalized Jain score rises from about 0.7741 for the welfare LP to about 0.8187 for the five-percent Nash configuration. Proportional allocation has a higher reported score of about 0.8642, while generating lower total saving. Therefore the Nash result cannot be described as simultaneously maximizing every fairness statistic and total welfare. It represents one explicit preference between economic value and a regularized benefit objective, evaluated with additional descriptive metrics.

Normalization is part of that interpretation. The pilot divides household benefit by a reference scale derived from available historical load and a fixed reference tariff. This makes the normalized quantity different from raw euros and different from benefit relative to income. A high normalized Jain score means equality in the selected scaled benefit, not a demonstrated social judgment about need or vulnerability. The reported scale is operationally reproducible, but its normative suitability for a real community would need agreement from participants and a clear explanation of why that scale was chosen.

The original aggregate Ridge forecast is separate from the first battery comparison. Section 6.11 now implements a distinct household-level Ridge predictor as an input to the same MPC and measures downstream cost. It uses a different target, input vector and sample from the aggregate pilot, so the two MAE tables cannot be compared directly. The learned IPPO/MAPPO actors still do not consume either forecast. This explicit connection establishes forecast-to-MPC evaluation without implying an unimplemented forecast-to-DRL pipeline.

The pilot retains 1,189 allocation intervals, whereas the battery experiment retains 704 intervals grouped into complete episodes. The difference follows from eligibility rules and should not be described as inconsistent measurement totals. An allocation decision can be evaluated at an isolated complete interval, while a battery episode requires a

continuous state trajectory and a preceding observation. Even within the same nominal June window, the two experiments therefore use different subsets. Comparing their absolute savings without accounting for these subsets would confound the intervention with data availability.

The pilot's primary role in the thesis is to establish the accounting and distributional problem, expose the common-tariff invariant and provide transparent allocation comparators. The battery extension then introduces an intertemporal state and tests whether learned control adds value under its declared protocol. The settlement experiment tests a third boundary: faithful recording of one fixed exchange stream. Keeping these three experiments separate makes their contributions complementary. Combining their best-looking metrics into one headline would imply an integrated system benefit that was never measured.

6 Battery control and learning results

6.1 Battery experiment protocol

The extension implements battery control, trained multi-agent policies and an executed settlement contract. Its result is negative for the current learning claim: under the fixed training budget, IPPO and MAPPO do not beat the causal MPC comparator. The contract succeeds as an accounting and authorization test. Neither finding establishes live trading performance, a general reinforcement-learning ranking or a blockchain business case.

Table 6.1 Battery experiment protocol

Element	Implemented specification
Battery per household	2 kWh nominal; 0.2–1.8 kWh bounds; 2 kW AC power; charge and discharge efficiency 95%
Episode	16 consecutive 15-minute intervals; all batteries start and finish at 1 kWh
Degradation	Assumed 0.01 EUR per AC kWh charged or discharged
Tariffs	Assumed import 0.20 before 07:00, 0.42 from 16:00 to 20:00, 0.30 otherwise; export 0.08; local fee 0.02 EUR/kWh
Splits	404 training episodes in January–April; 105 validation in May; 44 test episodes in June 1–14, 2020
Observation	Previous measured interval net energy, current storage, clock, remaining horizon, known tariff and household identity

Source: research implementation and recorded outputs. AI-assisted preparation.

Battery parameters are scenario assumptions, not measured StoreNet hardware specifications. The state transition applies charging and discharging efficiencies. A common feasibility guard limits requested AC energy by power, state of charge and reachability of the terminal target. The final interval restores the same initial energy; grid exchanges pay for restoration. No controller gains free energy from an end-of-episode reset.

Every episode is a complete aligned four-hour block with a valid preceding interval for the first forecast. Missing observations are not interpolated, and state never jumps across a gap. The 44 test blocks contain 704 intervals, 52.38 percent of the two-week window. Complete-block selection can bias time-of-day coverage; the results are not full-period electricity bills.

The original June study used the retained processed subset after an earlier raw-data recovery attempt returned HTTP 403. The 28 September acquisition succeeded, enabling the separate July–December evaluation in Sections 6.13–6.14. June 1–14 had already been inspected in the allocation pilot, so chronological training and validation do not make it confirmatory. The original controller totals were reproduced at the reported precision.

6.2 Learning and MPC implementation

IPPO and MAPPO are implemented with PyTorch and were actually trained (Yu et al., 2022; PyTorch, n.d.). Each uses the same parameter-sharing local actor: two 32-unit tanh hidden layers and five categorical action requests, −0.5, −0.25, 0, 0.25 and 0.5 kWh. Six households are agents with separate observations and actions, but share actor parameters. This is not a claim of six independently parameterized neural networks.

The controlled difference is the value critic during training. IPPO sees local observations; MAPPO sees all six current observations. Both receive the same mean community incremental savings reward relative to sharing with idle batteries. Thus the experiment tests centralized critic information, not selfish versus cooperative rewards. At execution, neither policy calls the critic or reads future demand.

Table 6.2 Learning and MPC implementation

Training setting	Value
Random seeds	11, 22, 33, 44 and 55 per algorithm
Budget per seed	60 updates × 12 sampled episodes × 16 joint steps = 11,520 environment steps
Household transitions	69,120 per seed; training episodes may repeat
PPO	4 full-batch epochs; clipping 0.2; learning rate 0.0003; entropy coefficient 0.01; gradient norm 0.5
Returns	Discount 1.0; GAE lambda 0.95; zero terminal bootstrap; normalized advantages
Checkpoint selection	Highest May validation savings at ten-update intervals; deterministic test actions

Source: research implementation and recorded outputs. AI-assisted preparation.

All requests pass the common physical guard. Multiple requests can map to the same executed action near a bound, making learning less efficient. The model checkpoints

and nonzero parameter changes show that optimization occurred; they do not establish convergence. The ten saved checkpoints permit evaluation without retraining.

Idle sharing leaves the batteries unused. The self-consumption rule responds to the previous measured net load. Causal community MPC repeatedly solves a linear program over the remaining episode with persistence net-load forecasts and the known tariff schedule. It includes storage dynamics, grid exchange, pooled local exchange, throughput cost and the terminal equality. Under these positive-price assumptions, simultaneous charge/discharge is unprofitable; every solution is checked before its first action is applied.

MPC has community information and continuous controls. The original learned actors have local observations and discrete requests. Section 6.9 extends the learned action representation under controlled conditions. Section 6.10 tests one fairness-shaped reward. Neither extension implements a daily risk constraint or confirms the original savings-retention hypothesis.

6.3 Controller outcomes

Costs below aggregate the same six households and 44 complete test episodes, including degradation. Positive savings means lower cost than idle-battery sharing. Learned-method means and sample standard deviations are across five seeds, not independent days.

Table 6.3 Controller outcomes

Method	Mean cost EUR	Seed SD EUR	Savings EUR
idle	159.4083	—	0.0000
self_consumption	171.1107	—	-11.7024
MPC	159.1604	—	0.2478
IPPO	173.3106	19.5128	-13.9023
MAPPO	182.8197	22.4156	-23.4114

Source: research implementation and recorded outputs. AI-assisted preparation.

MPC saves 0.2478 EUR against the 159.4083 EUR idle reference, approximately 0.16 percent. Self-consumption raises cost by 11.7024 EUR while cycling 412.7433 kWh, compared with 10.1645 kWh for MPC. Lag-based decisions, conversion losses and

terminal restoration are plausible explanations for the rule's weak result; their separate effects were not ablated.

IPPO averages 173.3106 EUR, with a range of 159.4083–206.7535 EUR. Two seeds execute no throughput and match idle sharing. MAPPO averages 182.8197 EUR, with a range of 162.6988–206.7535 EUR. The paired MAPPO-minus-IPPO cost difference averages +9.5091 EUR across five matching seed labels. No learning advantage was demonstrated. Larger budgets or redesigned actions could change this, but were not tuned on the test outcomes.

Physical feasibility and economic quality are separate findings. Maximum observed community energy residual is 8.88e-16 kWh; maximum terminal residual is 0.00e+00 kWh. Agents can satisfy these constraints and still make poor economic decisions. The physical model remains a single bus without feeder voltage or thermal constraints.

Household losses are retained. The chosen Jain interpretation is left undefined when savings contain negative values; all-zero incremental savings are not perfect fairness. Empirical 95 percent CVaR of episode costs is diagnostic: 44 four-hour episodes provide only 2.2 tail episode-equivalents. It is neither daily CVaR validation nor enforcement of a risk budget.

Five seeds measure optimization variation; they are not five independent communities. The same households recur across splits, blocks within a day can be dependent and tariff boundaries rarely fall inside selected four-hour blocks. The short horizons, missingness and small training budget limit conclusions about other seasons, communities or reinforcement learning in general.

6.4 Interpretation

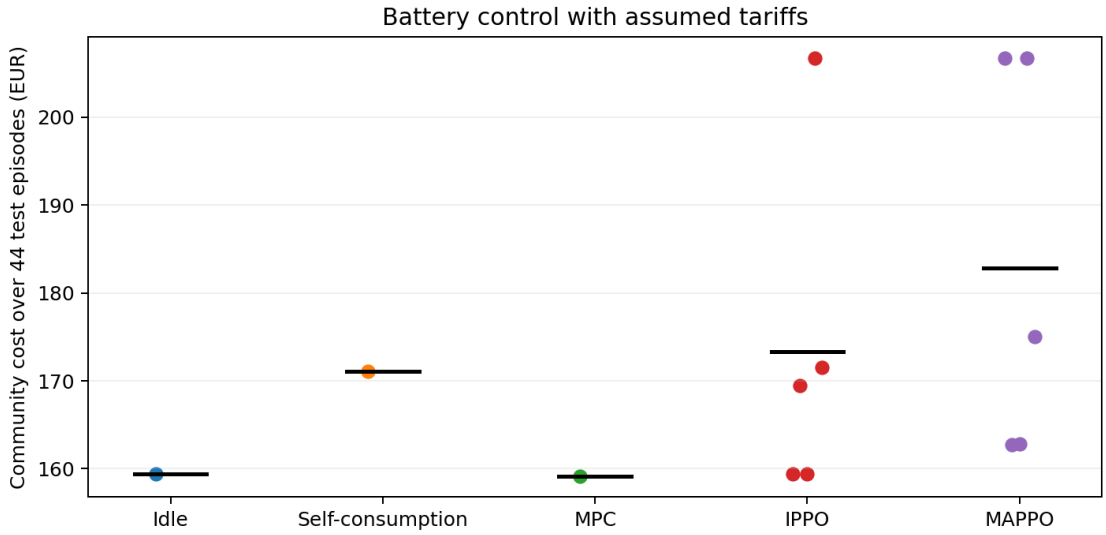


Figure 6.1 Each point is a controller seed; black horizontal marks show means. Lower cost is better. The common physical model is shared, while MPC has community observations and continuous controls. Source: research experiment outputs.

The allocation pilot compares distribution under static tariff scenarios. The original battery experiment holds proportional sharing fixed and tests storage under time-of-use tariffs. Sections 6.9–6.10 extend that battery design with explicit interventions. Allocation-pilot fairness gains are not evidence that the learned reward works. No 95 percent savings-retention guarantee has been demonstrated.

This is a useful negative thesis result: the implemented centralized critic did not outperform the local critic or causal MPC under a specified, reproducible budget. It supports retaining strong simple comparators and reporting failed learning runs. It does not justify a claim that reinforcement learning can never help energy communities. A follow-up must fix a new protocol before expanding training or inspecting new test data. The settlement layer is downstream of the controller. It records selected exchanges and checks accounting permissions; it does not improve a poor dispatch policy. Chapter 7 measures this distinction with a fixed stream from MAPPO seed 11 rather than selecting whichever seed gives attractive ledger outcomes.

6.5 Episode coverage and concentration of the MPC result

A post hoc audit of the saved household records adds a more specific explanation of the aggregate result. Community savings are summed over the six households within each episode before classifying the episode. Values within 0.000000001 EUR of zero are treated as numerical ties. MPC produces positive incremental savings in two of the

forty-four episodes and ties idle sharing in the other forty-two. It has no negative community episode under this tolerance. Thus the aggregate saving of 0.2478309 EUR is concentrated in two periods rather than distributed across the entire test sample.

The two positive episodes begin at 04:00 on 2 June and 3 June 2020. Their community savings are 0.1508630 EUR and 0.0969679 EUR, respectively. Together they account for the complete recorded MPC gain. These are descriptive decompositions of the original evaluation, not newly selected test cases. Reporting their dates allows a reviewer to locate the corresponding raw household records and verify that the sum matches the headline. It also prevents the average result from being interpreted as a benefit delivered in a typical four-hour window.

Eligibility is uneven across start times. The forty-four retained episodes comprise two starting at 04:00, eleven at 08:00, nine at 12:00, twelve at 16:00 and ten at 20:00. No 00:00 episode is retained. Each start-hour category had fourteen possible dates in the nominal two-week window. The available sample therefore covers only two of fourteen morning-transition blocks, while the 16:00 category retains twelve. These differences arise from the complete-block and preceding-observation rules; they do not establish why the underlying measurements are missing.

The interaction between episode boundaries and tariffs is central to interpretation. The import tariff rises from 0.20 to 0.30 EUR/kWh at 07:00, inside the 04:00–08:00 episode. In contrast, the rise to 0.42 at 16:00 and the fall back to 0.30 at 20:00 coincide with four-hour episode boundaries. Since every episode starts and ends at 1 kWh per battery, the evaluated controllers cannot buy energy in the preceding block and carry it into the 16:00 peak. This experiment consequently restricts intertemporal arbitrage opportunities that a continuous daily controller might exploit.

The concentration of recorded gains in the only retained 04:00 episodes is consistent with this tariff-and-horizon mechanism. It is not a causal decomposition proving that every euro of the gain comes exclusively from the 07:00 change: household net-energy patterns and local sharing also enter the simulation. A matched rerun with a constant tariff or a longer continuous horizon would be needed to isolate those effects. The current audit identifies a design limitation and a testable explanation without presenting either unexecuted ablation as a measured result.

The equal terminal-energy condition remains valuable because it prevents apparent savings from consuming a free initial battery inventory. Removing it without accounting for residual energy would create a different bias. The appropriate follow-up is a longer

physically linked simulation with a documented terminal value or restoration convention, not simply deleting the constraint. Twenty-four-hour episodes, for example, could include multiple tariff bands but would require enough complete data and a clear treatment of the final stored energy. Their results cannot be inferred by scaling the present four-hour totals.

Table 6.4 Retained episodes by start hour

Start hour	Retained	Possible	Coverage
00:00	0	14	0.0%
04:00	2	14	14.3%
08:00	11	14	78.6%
12:00	9	14	64.3%
16:00	12	14	85.7%
20:00	10	14	71.4%

Source: calculation from retained experiment records. AI-assisted preparation.

6.6 Training variability and unsuccessful control behavior

The five-run cost distribution provides more information than a single average. IPPO has median cost 169.4617 EUR, mean 173.3106 EUR and sample standard deviation 19.5128 EUR. MAPPO has median cost 175.0455 EUR, mean 182.8197 EUR and sample standard deviation 22.4156 EUR. The largest cost in both groups is 206.7535 EUR. The spread is substantial relative to the 0.2478 EUR improvement achieved by MPC. This makes the fixed training budget an important part of the result rather than an incidental configuration detail.

Two IPPO runs, seeds 11 and 33, have zero recorded battery throughput and exactly reproduce the idle-sharing aggregate cost. They are evaluated trained checkpoints, but their deterministic behavior does not provide an economic improvement. The records support calling them inactive on this test set. They do not by themselves establish why the policies became inactive, whether their sampled training behavior was also inactive, or whether a different state distribution would elicit different actions. Those questions would require action traces or additional evaluation.

The same numeric seed can be paired across IPPO and MAPPO to display a controlled bookkeeping comparison. MAPPO minus IPPO cost is +47.3452 EUR for seed 11, −6.7630 EUR for seed 22, +3.4390 EUR for seed 33, zero for seed 44 and +3.5243 EUR for seed 55. Positive differences favor IPPO. MAPPO is cheaper in one pair, more expensive in three and tied in one. The mean paired difference is +9.5091 EUR. Equal

seed labels do not guarantee identical random trajectories once model structure and sampling behavior differ, so this comparison is descriptive rather than a fully matched causal experiment.

A further audit counts episodes with positive or negative community saving. IPPO seed 22 loses in forty-one episodes and ties in three; seed 44 loses in all forty-four; seed 55 gains in two and loses in forty-two. MAPPO seeds 11 and 44 lose in all forty-four episodes. Seed 22 loses in twenty-two and ties in twenty-two; seed 33 loses in thirteen and ties in thirty-one; seed 55 gains in four, loses in thirty-two and ties in eight. Isolated favorable episodes therefore do not overturn the negative aggregate outcome for these policies.

Repeated aggregate costs should not automatically be interpreted as duplicate models. IPPO seed 44 and MAPPO seeds 11 and 44 have the same aggregate cost and throughput in the delivered records. Equal summaries can arise when different parameter values produce the same effective deterministic requests or when the projection collapses different requests to identical feasible actions. The present aggregate audit does not establish which mechanism applies. A claim about behavioral identity would require comparing the interval-level requests, projected exchanges and states, rather than relying on matching totals alone.

The original experiment cannot isolate why learning underperforms. Five discrete actions, two 32-unit hidden layers, sixty updates, information scope and horizon design are candidate factors. Section 6.9 now tests action representation and budget separately in a factorial comparison. Network size, information and horizon remain fixed, so the new study narrows the explanation without identifying a single universal cause.

Table 6.5 Paired seed cost differences

Seed	IPPO EUR	MAPPO EUR	MAPPO minus IPPO
11	159.4083	206.7535	+47.3452
22	169.4617	162.6988	-6.7630
33	159.4083	162.8473	+3.4390
44	206.7535	206.7535	+0.0000
55	171.5211	175.0455	+3.5243

Source: calculation from retained experiment records. AI-assisted preparation.

6.7 Household incidence and the participation constraint

The community objective hides a distributional distinction that becomes visible in the household totals. Under MPC, H1 loses 0.0257159 EUR relative to idle sharing. H2 gains 0.0386038, H3 gains 0.1050936, H4 gains 0.0340656, H5 gains 0.0040952 and H6 gains 0.0916886 EUR. Summing these signed values gives the community gain of 0.2478309 EUR. Negative values are retained in the results and are not set to zero before calculating the total. The small absolute amounts are consequences of the experiment's limited period and assumed tariffs; they are not annual household bill estimates.

This result distinguishes transaction-level sharing benefits from the incremental value of storage control. In the sharing calculation, a positive import-export spread can make a matched trade beneficial to buyer and seller relative to their grid-only exchange. Battery operation changes the household's exchange pattern and adds losses and throughput cost. It can therefore reduce a household's outcome relative to the stronger baseline of sharing with an idle battery. A participation argument proved for an individual sharing transaction does not automatically apply to a sequence of controlled battery actions.

The aggregate MPC surplus is arithmetically large enough to cover H1's recorded loss, but the experiment contains no compensation rule. Covering 0.0257159 EUR would consume approximately 10.38 percent of the total community gain. A transfer within the community would redistribute that gain while leaving its total unchanged before any transaction cost. This is a counterfactual accounting observation, not a simulated settlement policy. It cannot establish that members would accept the rule or that the rule would remain feasible in another period with a smaller or negative community surplus.

The other controllers reinforce the need to retain household-level outcomes. All six households lose under the self-consumption rule and under the three high-cost runs IPPO 44, MAPPO 11 and MAPPO 44. IPPO 22 has four losing households and IPPO 55 has five. MAPPO 22 and 33 each have four, while MAPPO 55 has six. A total saving or average reward should therefore be accompanied by the number of losing members and their signed monetary results. Applying an inequality index to a clipped benefit vector would erase precisely the losses that a participation rule must address.

A future fairness-constrained controller needs a declared outside option, settlement horizon and treatment of infeasible participation guarantees. It might require every household to remain no worse off over a complete day, or permit a bounded temporary loss compensated at the end of the period. Those designs imply different information

requirements and different obligations for the coordinator. The current prototype does not choose between them. Its contribution is to make the distributional issue observable and to show that optimizing the implemented team reward does not resolve it automatically.

Table 6.6 MPC household outcomes

Household	Cost EUR	Saving EUR	Throughput kWh
H1	8.6844	-0.0257	1.0395
H2	25.1545	+0.0386	1.7070
H3	19.6273	+0.1051	2.0895
H4	39.5740	+0.0341	1.9025
H5	9.0659	+0.0041	1.3107
H6	57.0544	+0.0917	2.1153

Source: calculation from retained experiment records. AI-assisted preparation.

6.8 Fixed dispatch sensitivity to the throughput charge

The recorded cost includes an explicit degradation proxy of 0.01 EUR for each kWh of absolute AC battery exchange. Because both charging and discharging contribute to throughput, the charge applies to both legs of a cycle. The stored results allow the arithmetic contribution of this proxy to be separated from the remaining cost without retraining a policy. This is a fixed-dispatch accounting sensitivity: the sequence of battery actions, realized exchanges and sharing allocations is held unchanged. It is not a new optimal policy for a different degradation assumption.

For a recorded controller with throughput Q and saving S at the original charge of 0.01 EUR/kWh, its saving at an alternative charge d is S plus (0.01 minus d) multiplied by Q, provided dispatch is held fixed. The same relationship can be read without any optimization: removing a recorded expense increases saving by that expense. For zero-throughput controllers the adjustment is zero. For positive throughput, dividing the saving excluding the recorded charge by Q gives the fee at which that fixed dispatch breaks even relative to idle sharing.

MPC has 10.1645 kWh of throughput and a recorded degradation charge of 0.1016451 EUR. Removing that charge while retaining the same dispatch raises its calculated saving from 0.2478309 to 0.3494761 EUR. Its fixed-dispatch break-even charge is approximately 0.0343820 EUR/kWh. At a charge of 0.02 EUR/kWh the same actions would save approximately 0.1461858 EUR; at 0.04 they would lose approximately 0.0571045 EUR. These values describe the accounting response of the saved dispatch.

MPC reoptimization at each fee could choose different actions and is not represented by this calculation.

The self-consumption rule has 412.7433 kWh of throughput, corresponding to 4.1274 EUR of the recorded charge. Even after removing that expense, its fixed-dispatch saving remains −7.5750 EUR. The worst-cost learned runs have 422.9558 kWh of throughput and a 4.2296 EUR charge; removing it leaves a saving of −43.1157 EUR. Thus the explicit degradation proxy alone does not explain their unfavorable result. The remaining gap reflects the net consequences of the recorded dispatch under conversion losses, tariffs, sharing and terminal restoration, without separately identifying each component.

Every active learned run has negative saving even after the recorded degradation charge is removed. Its fixed-dispatch break-even fee is therefore negative, which means no nonnegative throughput charge can make that unchanged behavior outperform idle sharing. This does not rule out a different policy trained under a zero-charge reward. Changing the reward changes the learning problem; holding the policy fixed changes only the accounting. Keeping those two questions distinct avoids presenting a cheap arithmetic sensitivity as a newly executed learning experiment.

The sensitivity is useful for engineering interpretation because it identifies which conclusions depend on the chosen expense convention. MPC's small positive margin depends on a sufficiently low charge for its particular saved dispatch. The losses of the active learned controllers persist even when the explicit charge is zero. Neither observation supplies a physical battery lifetime model or a warranty-based replacement cost. The 0.01 assumption remains a scenario parameter, and a deployment assessment would need a degradation model fitted to the intended battery chemistry, operating regime and financial decision.

Table 6.7 Fixed MPC dispatch sensitivity

Charge EUR per kWh	Saving EUR
0.000	0.349476
0.010	0.247831
0.020	0.146186
0.030	0.044541
0.040	-0.057105

Source: calculation from retained experiment records. AI-assisted preparation.

6.9 Controlled action and training budget study

A follow-up exploratory protocol was recorded before its runs on 21 September 2026. It crosses five discrete requests with bounded continuous requests and 60 with 300 training updates. Both IPPO and MAPPO use seeds 11, 22, 33, 44 and 55 in each cell, producing forty runs. The original data splits, twelve actor inputs, two 32-unit hidden layers, economic reward, physical projection and PPO update settings remain fixed. All cells use the same 44 June episodes; this is development evidence, not an independent test.

The continuous actor produces two positive Beta parameters using softplus plus one. A sample on the unit interval is shifted by minus 0.5 to produce an AC request between −0.5 and 0.5 kWh. Deterministic evaluation uses the Beta mean. The discrete policy retains the categorical argmax rule. Both use entropy coefficient 0.01, but discrete entropy and continuous differential entropy have different scales. The intervention therefore compares these specified policy parameterizations; it does not isolate abstract action resolution from every associated distribution choice (PyTorch, n.d.).

Training uses twelve episodes per update and sixteen joint decisions per episode. Sixty updates provide 11,520 joint steps, while 300 provide 57,600. Checkpoints are evaluated every ten updates on May community savings; strict improvement retains a new checkpoint and a tie keeps the earlier one. A larger budget also offers more validation candidates. The measured budget effect therefore includes this fixed selection procedure, rather than a comparison of arbitrary final iterates.

Table 6.8 Action and budget comparison

Variant	IPPO mean EUR	IPPO SD EUR	MAPPO mean EUR	MAPPO SD EUR
Discrete 60	173.3106	19.5128	182.8197	22.4156
Discrete 300	159.4083	0.0000	159.4083	0.0000
Continuous 60	173.3642	5.3303	175.1583	5.2994
Continuous 300	166.1422	2.4795	170.8108	3.9344

Source: sixty-run follow-up records, action subset. AI-assisted preparation.

Table 6.8 reports total cost over the same episodes and sample standard deviation across five training seeds. All ten discrete 300-update runs cost 159.4083 EUR with zero executed throughput, reproducing idle sharing. The mean reduction relative to discrete 60 updates is 13.9023 EUR for IPPO and 23.4114 EUR for MAPPO. Thus the original network and discrete action set can represent a non-cycling policy, and the larger-

budget procedure removes the recorded excess cost in this setting. It still misses MPC's 0.2478 EUR saving.

For IPPO, the continuous mean cost changes from 173.3642 EUR at 60 updates to 166.1422 EUR at 300, a change of -7.2220 EUR. At 60 updates, continuous minus discrete cost is +0.0537 EUR; at 300 updates it is +6.7339 EUR. These conditional contrasts show why changing action type and budget together would obscure their separate effects. They are descriptive averages, not confidence bounds for unseen periods.

For MAPPO, the continuous mean cost changes from 175.1583 EUR at 60 updates to 170.8108 EUR at 300, a change of -4.3475 EUR. At 60 updates, continuous minus discrete cost is -7.6614 EUR; at 300 updates it is +11.4025 EUR. These conditional contrasts show why changing action type and budget together would obscure their separate effects. They are descriptive averages, not confidence bounds for unseen periods.

Every follow-up run retains its selected checkpoint, full validation trajectory, aggregate request magnitude, execution-projection rate and household outcomes. Projection can map different requests to the same feasible action, especially at the terminal boundary. Zero executed throughput therefore describes realized behavior on this evaluation set, not necessarily zero raw requests everywhere. The available diagnostics do not establish that a single failure mechanism explains every seed.

6.10 Controlled fairness reward study

Twenty further runs compare economic reward alone with a fixed distributional penalty: two algorithms, two reward arms and five seeds. Both arms retain discrete actions, sixty updates and two 32-unit hidden layers. Their observation adds the household's own cumulative realized saving and the community mean to the original twelve inputs. This gives fourteen components to both actors and 84 to the centralized critic. The community mean is an additional shared aggregate, so these branches are compared with each other rather than interpreted as a reward-only change from the original twelve-input models.

Let cumulative household savings be measured relative to idle sharing in EUR. The potential is the negative population variance across the six cumulative savings, after division by a fixed 1 EUR reference. It is defined for negative savings. The common reward is mean incremental community saving plus 0.1 EUR times the change in that

dimensionless potential. The reference and weight are modeling choices fixed before these runs, not parameters selected from June results. With discount factor one, the potential increments telescope to the terminal penalty. The terminal term is retained deliberately, so the objective changes; no policy-invariance claim is made.

Equal losses have zero dispersion but remain economically undesirable. The economic term is therefore essential, and signed monetary outcomes accompany the dispersion score. Both arms use the same May economic checkpoint rule, isolating the training-reward intervention within this augmented state. This study does not optimize a fairness-based checkpoint rule or search over penalty weights. Reported dispersion is the mean across episodes of the within-episode household saving variance in EUR squared; it is not Jain, daily fairness or pooled-total variance.

Table 6.9 Fairness reward comparison

Algorithm and reward	Mean cost EUR	Seed SD EUR	Mean variance EUR²	Mean losing homes
IPPO Economic	178.1258	25.5314	0.001451	2.6
IPPO With penalty	178.0594	25.3963	0.001484	3.0
MAPPO Economic	168.1648	15.8780	0.001780	2.0
MAPPO With penalty	168.6638	15.7868	0.002156	2.4

Source: matched fourteen-input follow-up records. AI-assisted preparation.

For IPPO, adding the penalty changes mean cost by -0.0665 EUR and mean episode dispersion by +0.000033 EUR squared. Cost is lower in 2 of five paired seeds, higher in 1, and tied in 2. The mean number of households with negative pooled savings changes from 2.6 to 3.0. The archive preserves the individual runs so these averages cannot hide a losing seed or household.

For MAPPO, adding the penalty changes mean cost by +0.4990 EUR and mean episode dispersion by +0.000376 EUR squared. Cost is lower in 1 of five paired seeds, higher in 2, and tied in 2. The mean number of households with negative pooled savings changes from 2.0 to 2.4. The archive preserves the individual runs so these averages cannot hide a losing seed or household.

The result answers a narrower question than whether fairness-aware reinforcement learning succeeds in general. It measures one signed-benefit penalty, normalization, information state, training budget and checkpoint rule on already inspected dates. A favorable change in one metric cannot establish universal improvement, individual

rationality or the proposed 95 percent savings-retention criterion. Independent data and development-only selection of alternative weights remain necessary.

The evidence package contains sixty follow-up checkpoints and 15,840 household-episode rows, alongside the ten original models. These rows repeat the same six households and forty-four periods; they are not new independent observations. Four additional tests verify the potential's signed-value and telescoping behavior, causal extra state, finite continuous-policy gradients and physical feasibility. The 28-test suite for those modules passed; the final suite including market and coupling checks contains 37 tests. Evaluation-only replay is checked against the retained outputs and original-reference outcomes.

6.11 Forecast substitution and downstream control cost

A further exploratory experiment connects forecast output to an actual control decision. The controller, physical model, episode set, tariffs and proportional settlement remain fixed while the remaining-horizon household net-load forecast changes. This directly evaluates one coupling that was absent from the original allocation pilot. The four arms use last-value persistence, previous-day persistence, recursive Ridge and perfect future measurements. The last arm is explicitly non-causal and is used only as an information benchmark.

The Ridge model predicts six household net-energy values. Its inputs are the previous interval for all six households, the corresponding previous-day values and sine/cosine clock terms. Standardization and Ridge with alpha equal to one are fitted only on 9,442 complete January–April rows. The configuration is recorded before evaluation and is not selected from June performance. A remaining horizon is generated recursively, substituting predicted values into subsequent lag-one inputs. The previous-day values remain historical observations.

Every forecast call truncates the observation history strictly before the current decision. If any household is missing at a previous-day lag, the complete six-household lag vector falls back to the last observed interval. Both seasonal and Ridge arms apply the same rule. It is invoked for 473 of 5,984 forecast horizon slots, about 7.9 percent. Missing source observations are not filled into the realized target. All four arms retain exactly the same 44 complete evaluation episodes.

Table 6.10 Forecast substitution with a fixed MPC

Forecast and controller	Cost EUR	First-step MAE kWh	Horizon MAE kWh	AC throughput kWh
Persistence MPC	159.1604	0.08295	0.14948	10.165
Previous-day MPC	160.3051	0.17319	0.18046	76.449
Ridge MPC	158.8618	0.08424	0.12770	30.827
Oracle MPC	153.7614	0.00000	0.00000	77.075

Source: forecast control records. AI-assisted preparation.

First-step MAE covers 4,224 household predictions at 704 actual control decisions. Horizon MAE covers 35,904 prediction occurrences because overlapping horizons repeatedly forecast the same future intervals. These are different weighting schemes, and neither count supplies additional independent observations. The target is per-household net energy rather than the aggregate target used in Chapter 5. All numerical comparisons in Table 6.10 share the same realized outcomes.

Ridge MPC reduces total cost by 0.2986 EUR, or about 0.188 percent, relative to persistence MPC. It has slightly worse first-step MAE but lower remaining-horizon MAE. Relative episode cost is lower in thirteen cases, tied in twenty and higher in eleven. This is a measured example in which one-step prediction accuracy alone fails to rank downstream control value. It is not a significance claim or evidence that arbitrary improvements in forecast accuracy cause greater profit.

Distribution remains material. Ridge MPC saves 0.5464 EUR in total against idle sharing, yet H1 loses 0.0578 EUR and H5 loses 0.0138 EUR. Previous-day MPC loses 0.8968 EUR in total. The perfect-information arm costs 153.7614 EUR, leaving a 5.1005 EUR gap to Ridge MPC under this model. The same solver is retained, but the gap describes this information intervention and implementation; it is not a value estimate for an attainable forecast or a Polish trading strategy.

Energy-balance residuals in all four arms remain at or below 4.45×10^{-16} kWh and terminal residuals are zero. The unchanged persistence arm reproduces the original MPC total. A causality test alters every present and future measurement and verifies that each causal forecast is unchanged. The full research suite now contains 37 passing tests. Forecast conditioning of IPPO/MAPPO, a learned price forecast and a new independent evaluation period remain separate unfinished work.

6.12 Temporal sensitivity of the June forecast result

Ridge MPC reduces the June total cost by 0.2986 EUR relative to persistence MPC. Aggregating the paired contrasts across the six households within each of fourteen source days gives a mean difference of -0.02133 EUR per retained day. This refers to the included episodes within each day, not a complete daily bill. Ridge is cheaper on 9 days, tied on 1 and more expensive on 4.

Table 6.11 June temporal sensitivity

Block length in days	Lower EUR per day	Upper EUR per day
1	-0.04786	0.00020
2	-0.04668	0.00067
3	-0.04737	0.00096
7	-0.03925	-0.00341

Source: recomputed retained experiment records. AI-assisted analysis and table preparation.

Intervals with block lengths of one, two and three days include zero. The seven-day interval excludes zero, but fourteen observed days supply very few distinct week-length blocks. Choosing that result alone would exaggerate certainty. Leaving out any one day preserves a negative mean contrast, yet does not restore independence or remove temporal dependence. The magnitude is small and remains an exploratory finding.

6.13 Coverage and frozen model results in July to December

The wider period contains 184 calendar days and 1,104 candidate four-hour windows. The complete-case procedure retains 501 episodes across 168 source days, or 45.38% of candidate windows. The resulting 8,016 common fifteen-minute intervals extend seasonal coverage substantially but do not constitute 184 complete daily trajectories. Table 6.12 reports the monthly selection pattern.

Table 6.12 Later period coverage

Month in 2020	Eligible blocks	Candidate blocks	Coverage percent
2020-07	80	186	43.01
2020-08	91	186	48.92
2020-09	80	180	44.44
2020-10	74	186	39.78
2020-11	77	180	42.78
2020-12	99	186	53.23

Source: recomputed retained experiment records. AI-assisted analysis and table preparation.

Table 6.13 gives total community cost on the same retained episodes. Learned rows are means over five seeds, with their sample standard deviation. A positive saving denotes a lower cost than idle sharing. Every controller faces the same measured sequence and physical accounting; the table does not imply identical decision information across MPC and decentralized actors.

Table 6.13 Later period controller outcomes

Controller	Cost EUR	Seed SD EUR	Saving vs idle EUR
Idle sharing	1947.4997	—	0.0000
Self-consumption rule	2065.3425	—	-117.8428
Persistence MPC	1946.6754	—	0.8243
Previous-day MPC	1966.2359	—	-18.7361
Ridge MPC	1939.3432	—	8.1565
discrete 60 IPPO	2096.0545	208.2966	-148.5548
discrete 60 MAPPO	2198.6065	238.2748	-251.1067
discrete 300 IPPO	1947.4997	0.0000	0.0000
discrete 300 MAPPO	1947.4997	0.0000	0.0000
continuous 60 IPPO	2095.7490	54.3205	-148.2493
continuous 60 MAPPO	2114.1018	52.6341	-166.6021
continuous 300 IPPO	2031.5809	28.9311	-84.0812
continuous 300 MAPPO	2082.3081	42.5518	-134.8084
Matched control IPPO	2146.7170	272.3655	-199.2173
Matched control MAPPO	2034.0935	160.3715	-86.5938
Fairness λ 0.1 IPPO	2146.0237	270.5777	-198.5239
Fairness λ 0.1 MAPPO	2039.5124	160.0499	-92.0126

Source: recomputed retained experiment records. AI-assisted analysis and table preparation.

6.14 Temporal transfer and statistical interpretation

Coverage is strongly dependent on start time: only 1 of 184 midnight windows and 11 of 184 windows starting at 04:00 are eligible, compared with 120–124 windows at each daytime/evening start. The 04:00 block contains the 07:00 tariff transition, so selection removes most opportunities for that type of arbitrage. Daytime complete-case evidence cannot be treated as a representative full-day energy-storage assessment. Figure 6.2 pairs this coverage pattern with monthly forecast-control contrasts.

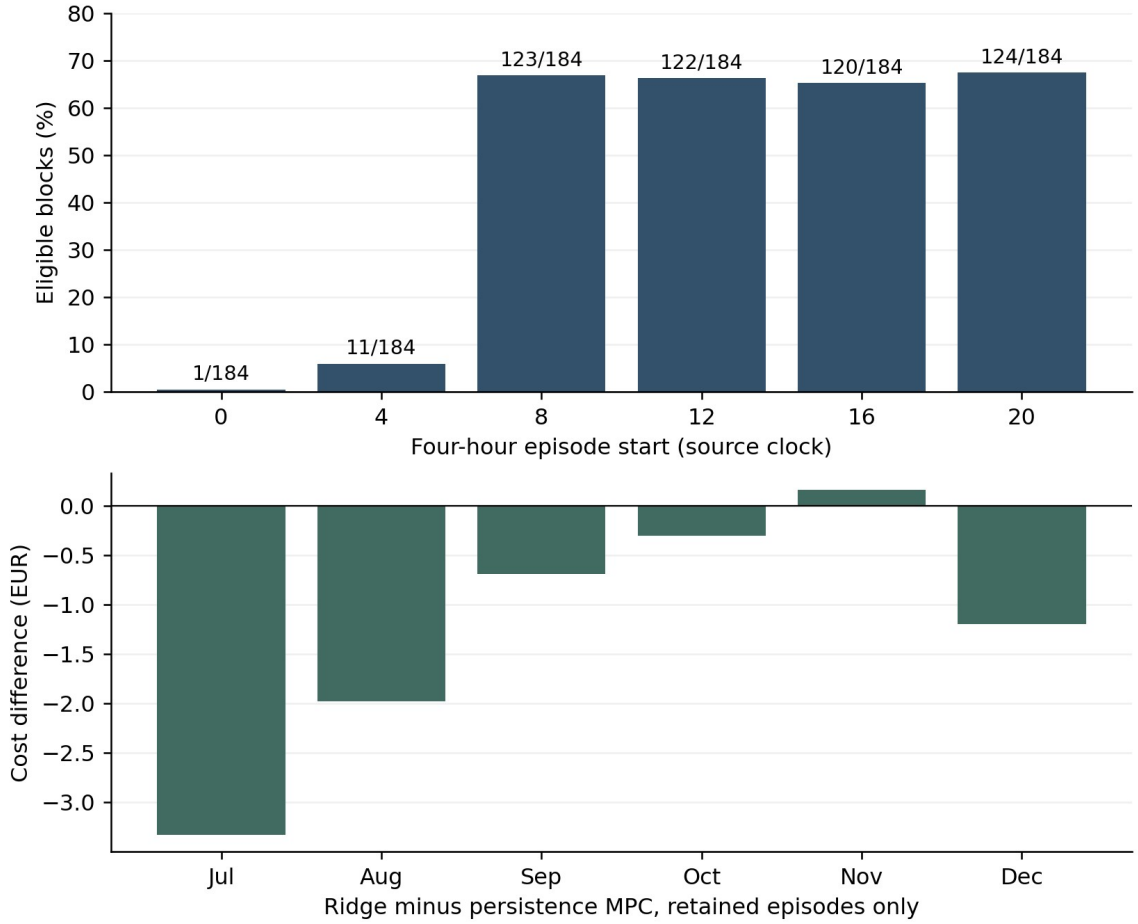


Figure 6.2 Coverage by start hour and monthly Ridge minus persistence cost

Source: retained later-period eligibility and cost records. AI-assisted analysis and figure preparation.

The primary discrete-IPPO contrast is +0.8243 EUR in total, or +0.00491 EUR per retained source day, against persistence MPC. Its seven-day block interval is [-0.00008, +0.01111] EUR per retained day. The sign convention is controller cost minus comparator cost; negative values favor the first controller. This uncertainty analysis is conditional and exploratory, as specified in Section 4.7.

Ridge minus persistence MPC is -7.3322 EUR in total and -0.04364 EUR per retained day. The seven-day interval is [-0.06889, -0.02127]. This is a temporal transfer check of the already fitted forecast, with no later-period refitting. A change in ranking therefore provides evidence about the stability of that fixed pipeline under later profiles; it does not isolate weather, missingness, forecast drift or individual model mechanisms.

The reward intervention also fails to improve the intended dispersion metric in the later period. Averaging episode-level population variance of signed household savings over episodes and seeds gives 0.001493 EUR² for the matched IPPO control and 0.001526 EUR² with the penalty. For MAPPO the corresponding values are 0.001737 and 0.002006 EUR². The small IPPO cost reduction under the penalty therefore

accompanies higher dispersion, while MAPPO worsens on both measures. These paired interventions describe one weight and selection rule; they do not rule out every fairness-aware policy.

The primary seven-day IPPO interval includes zero, and its point estimate favors persistence MPC by less than one euro over all retained episodes. This supports neither learning superiority nor equivalence. Ridge retains a negative cost contrast at every reported block length, yet the reduction is about 0.38% of persistence cost and remains conditional on the selected observations. It is cheaper on 60 retained days, tied on 54 and more expensive on 54; its aggregate advantage is not a daily guarantee.

Table 6.14 Later period household savings against idle

Household	Persistence MPC EUR	Ridge MPC EUR
H1	-0.3434	0.7273
H2	0.0502	1.2966
H3	0.3988	1.1444
H4	0.4772	2.3516
H5	-0.2314	0.9197
H6	0.4729	1.7169

Source: recomputed retained experiment records. AI-assisted analysis and table preparation.

Ridge has positive aggregate savings for all six households over this retained later period, whereas persistence MPC leaves H1 and H5 with losses. This differs from the June incidence and is not an enforced participation guarantee. A household can still lose during individual episodes. The common reward and fairness penalty impose no hard nonnegative-benefit constraint.

Table 6.15 Conditional paired daily cost contrasts

Contrast	Mean EUR per day	Seven day lower	Seven day upper
Discrete IPPO minus persistence	+0.00491	-0.00008	+0.01111
Ridge minus persistence	-0.04364	-0.06889	-0.02127
IPPO penalty minus control	-0.00413	-0.00522	-0.00309
MAPPO penalty minus control	+0.03226	+0.02615	+0.03851

Source: recomputed retained experiment records. AI-assisted analysis and table preparation.

All 65 controller and seed arms completed with maximum energy residual 1.33e-15 kWh and maximum terminal-energy residual 0.00e+00 kWh. The archive retains the

outcomes for every household, seed and episode, including losses. There are 11 gaps longer than one day between retained dates, with maximum spacing 6 days. Block-length sensitivities and monthly contrasts are included in the result files; the intervals are not adjusted for the secondary comparisons.

The extension narrows one important evidence gap by testing the same frozen implementation over a longer, later period. It leaves household sampling, incomplete coverage, four-hour inventory resets and scenario tariffs unchanged. It also leaves the independent-confirmation gap open because of the documented prior access. Neither a narrower interval nor more fifteen-minute rows can resolve those design limits. A new externally held period, linked longer horizons and verified market inputs answer different remaining questions.

7 Settlement experiment

7.1 Local contract and signed database

EnergySettlement was compiled with Solidity 0.8.30 for the Prague EVM and deployed to a local eth-tester/PyEVM chain using web3.py (Solidity, n.d.; Ethereum.org, n.d.; web3.py, n.d.). Six registered households receive equal accounting credits. A designated operator submits a unique identifier, participants, micro-kWh, micro-EUR proceeds and fees. The contract checks authorization, replay, membership, positive amounts and buyer credit before updating buyer, seller and treasury atomically.

The first 64 positive test trades from the fixed MAPPO seed 11 checkpoint were replayed both on the contract and in an Ed25519-signed SQLite transaction log. The two systems received identical integer inputs. Credits are research accounting units; no real currency, externally funded wallet or physical electricity transfer occurred.

Table 7.1 Local contract and signed database

Measurement	Observed result
Replayed trades	64
Final balances	Exact parity for six households and fee account
Credit conservation	6,000,000,000 micro-EUR accounting units
Deployment gas	641,807
Total settlement gas	4,424,772
Median settlement gas	68,917
Median local EVM call	21.457 ms
Median signed SQLite commit	0.217 ms

Source: research implementation and recorded outputs. AI-assisted preparation.

The contract rejected duplicate identifiers, unauthorized calls, insufficient credit, self-trades, zero energy and zero amounts. Failed calls preserved balances and did not consume the identifier. SQLite rejected duplicate records and insufficient balances with rollback; signature verification detected changed payload bytes. All ten ledger checks passed.

Timing is one sequential local pass and includes Python execution and receipt handling. It is not public-chain throughput, distributed consensus latency or finality. Gas units

were measured; no fiat gas-price assumption was added. An operator can still submit false metering to either system. Database verification requires an independently trusted public key; a key stored alongside a database does not by itself establish trust.

7.2 Reproducibility and failure semantics

All 64 accepted records conserve total credit and yield identical balances in both implementations. Failed contract calls revert atomically. A rejected identifier remains available for a later valid request, so failure does not silently consume an otherwise valid settlement identifier. The database counterpart rolls back an overspending transaction. These properties are local state-machine requirements, not a proof of distributed consensus or resistance to false meter reports.

The operator defines the transaction stream and remains a trusted source of metering and allocation information. A signature proves control of a signing key over recorded bytes; it does not establish the truth of the underlying electricity measurement. A public key delivered beside a database must be authenticated separately before an independent verifier can attach identity to it. No secret signing key is included in the package.

7.3 Transaction semantics and the boundary of the settlement claim

The ledger experiment starts from one fixed sequence of positive local exchanges generated by the MAPPO seed 11 evaluation. Selecting that sequence does not endorse the policy economically. It provides an identical input for comparing accounting implementations even though the controller performs poorly on the cost objective. This separation is useful: the ledger can be correct while the decisions it records are uneconomic. A successful settlement test cannot be used as evidence that the controller improves household welfare.

Each record identifies a seller, buyer, interval, energy quantity, seller amount, fee and unique identifier. Energy is rounded to micro-kWh and accounting amounts to integer micro-units. The buyer balance is reduced by the seller amount plus the fee, the seller balance increases by the seller amount, and the fee account increases by the fee. Those three updates sum to zero. Six household accounts begin with one billion units each and the fee account begins at zero, making the conserved total six billion units. These are test credits rather than deposits, bank balances or transferable legal claims.

The use of integer accounting makes exact equality meaningful within the ledger comparison. Both systems consume the same rounded records, so equal balances establish agreement on that integer stream. This is different from claiming exact

equality with unrounded floating-point energy economics. Each conversion to micro-units can introduce rounding, and a complete accounting system would need a documented residual policy. The current experiment validates its retained integer records and does not claim that rounding disappears or that the ledger independently reconstructs the full simulation.

A transaction identifier is derived from the serialized source record. The contract refuses to process the same identifier twice. The database enforces uniqueness on its identifier column and updates balances within a transaction. These mechanisms address duplicate submission for a known identifier. They do not by themselves recognize two different identifiers that describe the same physical meter event. Detecting semantic duplicates would require a canonical event key and rules for corrected or resubmitted measurements. Such rules belong to the metering and settlement specification, not merely to the choice of ledger technology.

The contract's authorized operator supplies the amounts and energy. The experiment verifies authorization and selected rejection paths, but it does not make the contract an independent electricity-price oracle. A syntactically valid amount can still originate from an incorrect meter reading or an inappropriate allocation rule. In the database branch, a valid digital signature demonstrates that the payload was signed by the corresponding key. Trust in the meter, in the operator's authority and in the association between that key and the operator remains external to the signature check.

7.4 Reproducibility and independent checking of the stored ledger

Reproducibility has several levels in this package. A file-hash check detects whether a delivered artifact differs from the recorded version. Recomputing totals from household rows checks internal numerical consistency. Evaluating saved model checkpoints reruns the control decisions without repeating training. Repeating training tests the wider training procedure and can produce different bytes or timings even when the protocol is retained. These operations answer different questions and should not all be summarized as a single claim that the experiment was reproduced.

The signed SQLite file and its public key permit an additional read-only check of the retained ledger. A verifier can open the database without modifying it, verify each signature over the exact stored payload bytes, compare the decoded payloads against the retained settlement stream, reconstruct the account updates and compare the reconstructed totals with the stored balances. This checks the internal agreement of the

delivered evidence. It still cannot establish the identity of the operator unless the public key is authenticated through an independent channel.

The stored database contains signatures rather than the private signing key. Re-running the full experiment generates a new signing key and a new database. Consequently, signatures and database bytes need not match across independent runs even if transaction amounts and final balances do. The logical outcome should be compared separately from byte identity. The existing hash manifest identifies the delivered run; it is not a requirement that every fresh run reproduce incidental metadata or cryptographic randomness.

Gas usage and elapsed local time also have different evidential meanings. Gas measures execution under the compiled contract and test environment, while wall-clock measurements include local processing overhead. A one-pass median of approximately 21.46 ms for the local EVM and 0.217 ms for SQLite is not a benchmark of public-network finality, adversarial consensus or distributed availability. The measurement supports describing the local implementations, but it cannot support a deployment throughput target or a public-chain operating budget without a separate benchmark.

The additional read-only verification was executed on the delivered database. All 64 stored signatures verified against the delivered public key; decoded payloads and identifiers matched the retained stream, and reconstructed balances matched both SQLite and the saved ledger summary. The conserved total was 6,000,000,000 accounting units. The script and its output are included as scripts/verify_saved_ledger.py and results/extension/saved_ledger_verification.json. This result verifies the retained artifacts without re-running an EVM or establishing the external identity of the signer.

8 Innovation and implementation potential

8.1 Nature of the contribution

GridMind combines established methods in an executable research case. Linear programming, PPO, centralized value functions, digital signatures and Ethereum accounting are not novel algorithms introduced by this thesis. The contribution lies in the tested integration, the matched comparison between local and centralized critics, the preservation of household losses, and the separation of energy decisions from transaction recording. This is a system and evaluation contribution rather than a claim of patentable novelty.

The allocation pilot exposes a structural invariant: with common buy, sell and fee parameters, total savings depend on matched volume. An allocation comparison that omits this invariant can overstate the economic importance of a redistribution rule. The battery extension then creates an intertemporal decision problem while holding local sharing fixed. Together these experiments make the research question more precise: economic opportunity must exist before a more complex controller can capture it.

The observed learning failure is informative because the implementation records checkpoint selection, multiple seeds, causal observations and terminal feasibility. It does not establish that reinforcement learning is generally unsuitable for energy management. A different dataset, horizon, observation set or training budget may reverse the ranking. Any such follow-up needs a new protocol and data that have not already informed methodological choices.

8.2 Potential users and decision process

The immediate user is an analyst or coordinator evaluating energy-community operating rules. The prototype supports comparative simulation before a physical or financial commitment. Household members are the beneficiaries whose costs and losses must remain visible. A technology vendor could use the reproducibility structure to compare a proposed controller with a transparent baseline, provided that customer data, operational constraints and governance requirements are added separately.

An implementation decision would begin with reliable metering, known time semantics and an agreed definition of the outside option. The next decision concerns whether the opportunity comes from local sharing, temporal arbitrage or participation in an external market. These revenues must be represented separately to avoid double counting. Controller selection follows after the relevant constraints and information times have

been fixed. Ledger choice follows from the required trust arrangement rather than from the presence of artificial intelligence.

8.3 Economic interpretation and investment boundary

The original June persistence-MPC saving against idle is 0.2478 EUR across six households and 44 selected four-hour episodes. It cannot support an annual saving or payback estimate because the retained blocks cover only 52.38 percent of the two-week period and do not form a representative annual sample. Multiplying the result by the number of days in a year would assume away seasonality, data selection and operating differences. The experiment provides no installed hardware price, maintenance budget or observed financing rate.

An investment model would require verified acquisition and installation costs, capacity replacement, operating expenditure and a cash-flow profile over a stated horizon. Benefits would include only revenues attributable to the chosen intervention after network charges, fees and degradation. Discounted cash flows would then be compared with the initial investment. The present evidence is insufficient to populate that model, so a positive net present value is not claimed.

Distribution also matters. The recorded MPC community saving coexists with a minimum household incremental benefit of approximately −0.0257 EUR. A coordinator that adopts the policy solely on total savings would overlook this loss. A future participation rule could require consent, compensation or a constrained allocation, but such a rule must be specified and evaluated before it is called individually rational. No compensation mechanism is simulated in this experiment.

8.4 Ledger choice and operational responsibility

The signed-database and contract comparison demonstrates that both systems can record the tested allocations correctly. Under the retained assumption of one trusted operator, the local evidence does not establish an additional commercial benefit from blockchain. A different trust model with independent parties may justify a shared ledger, but that would also require a defined consensus protocol, membership rules, key recovery, corrections and dispute handling. The local timing result cannot price these organizational requirements.

A production system would need procedures for missing or revised meter data, failed communications, controller fallback and settlement correction. The retained idle controller provides a transparent research comparator but is not a certified operational

fallback. Correction records should preserve the original event rather than silently overwrite history. Responsibility for data accuracy and dispute resolution would need to be agreed outside the numerical model.

8.5 Conditions for a follow-up pilot

A defensible next study would freeze its economic scenario and evaluation plan before inspecting new outcomes. It would use a longer period with verified time stamps, compare controllers under matched information where possible, and separate the effect of observations, action discretization and training budget. It would report raw household benefits, losses, energy residuals and meaningful uncertainty rather than rely on one aggregate score.

Section 6.10 implements a fairness-reward intervention separately from the economic action/budget study. A daily risk constraint remains a distinct unimplemented extension. It would need complete daily outcomes, a declared loss budget and adequate tail coverage. The present episode-level CVaR diagnostic cannot validate that stronger claim.

8.6 Engineering management decision criteria

An engineering management decision begins with the service being evaluated. In this prototype the service is coordinated local energy sharing with optional household battery control and traceable internal accounting. The coordinator's implemented optimization target is community operating cost, while households experience individual costs and losses. The prototype does not represent a supplier bidding in a wholesale auction, an aggregator selling reserve capacity or a financial trader holding market positions. Keeping that target precise links the software evaluation to a decision that a community coordinator could actually investigate.

The evidence suggests a sequence of decision gates rather than immediate deployment. The first gate concerns data: are measurement units, timestamps, completeness and household membership sufficiently understood to support the intended comparison? The second concerns economics: does a transparent baseline identify an opportunity after relevant costs? The third concerns distribution: who gains, who loses and what participation rule is acceptable? The fourth concerns control reliability: can the chosen policy meet physical constraints and recover from missing or delayed inputs? Only after these questions are answered does the choice of record-keeping architecture become decisive.

The current package passes a limited research gate: it contains executable implementations, measured input profiles, retained model checkpoints and internally consistent results. It does not pass an investment gate because the small exploratory improvement cannot establish annual cash flow or installed-system value. It does not pass a field-operation gate because communication delays, feeder limits, meter failures and device interfaces are outside the executed model. This distinction prevents the maturity of the code artifact from being mistaken for the maturity of a deployable energy service.

A manager comparing alternatives should use the same service boundary for their costs and benefits. If a battery is already installed for backup, the incremental control decision differs from a decision to buy a new battery solely for trading. If a database already meets the operator's audit requirements, the incremental benefit of a ledger must be identified separately from the benefit of energy optimization. The present results do not contain the capital and organizational costs required to compare those cases numerically. They provide a structured way to identify which additional measurements would change the decision.

A useful follow-up pilot would assign ownership of each evidence stream. The energy workstream would verify units, tariff applicability, battery specifications and operational constraints. The software workstream would retain data versions, control logic, test results and reproducible execution instructions. The finance workstream would define the outside option, cost categories, allocation of benefits and investment horizon. These are proposed responsibilities for a future pilot, not a certified record of what each student has already authored. The final team contribution declaration must describe actual reviewed work.

8.7 Remaining experiments and independent validation

The next priority is independent evaluation. June 1–14 remains development evidence after both the initial pilot and subsequent interventions. A new period requires an access history and fixed dates, eligibility, tariffs, horizons, metrics and selection rules before outcomes are opened. A later period is not automatically untouched. The 180-day target is a coverage goal, not a power justification, and it supplies only nine tail-day equivalents at 95 percent.

The reward and action/budget studies have now been executed on the retained period. They should be replicated on fresh data before their rankings guide a deployment

decision. The fairness experiment uses one fixed penalty weight and a common economic validation rule; it does not exhaust possible trade-offs or select an optimal fairness controller. Weight selection and any revised checkpoint objective belong to development validation before final testing.

Network width, observation scope and episode horizon remain separate interventions. The observed benefit of a larger discrete-policy budget cannot establish that the original network is optimal, nor that continuous control is intrinsically inferior. The continuous arm changes the distribution, deterministic action rule and entropy scale. Further comparisons must identify which implementation choices are being held fixed and which are part of the intervention.

A longer continuous horizon should permit storage transfers across relevant tariff changes while keeping initial and terminal inventory accounting consistent. Data exclusions must be reported by clock time and season. Inference should separate variation among training seeds from temporal uncertainty and household sampling. A future procedure needs a development-data precision assessment and a prespecified analysis appropriate to dependent observations.

The delivered protocols distinguish the executed exploratory studies, including the later temporal extension, from remaining confirmatory work. No untouched test is certified. Ledger parity remains separate: matching signed SQLite and local EVM balances does not establish meter truth, public-network resilience or a commercial advantage.

8.8 Evidence based communication of the contribution

The contribution is strongest when stated as an auditable evaluation with an explicit negative learning result. The prototype implements the physical and accounting mechanics, trains multiple policies, compares them with transparent alternatives and records where the expected advantage does not appear. This is more informative than treating the presence of reinforcement learning or a smart contract as an achievement independent of measured outcomes. The evidence establishes what the delivered system does under stated assumptions; broader effectiveness remains an empirical question.

The defense separates modeled physical correctness, controller performance in each evaluation period, household incidence and ledger agreement. June and July–December results must be identified explicitly; a June ranking cannot be stated as a universal result. Each claim has its own supporting artifact. The findings do not estimate annual profitability, universal fairness or live-grid readiness.

This communication also supports a practical research handover. A reviewer can identify the exact input, reproduce the descriptive analysis, inspect household losses, evaluate saved policies and verify the stored ledger. The next researcher can choose a specific unresolved mechanism rather than reconstructing the experiment from a presentation alone. The management value lies in reducing uncertainty about what has been tested, what failed and which next experiment would change the decision. It does not depend on presenting every technical component as novel.

8.9 Requirements for a reviewable pilot handover

A reviewable handover should let another person identify the question, reproduce the relevant evidence and understand the conditions under which a decision may change. For GridMind, the package already supplies the processed input, scenario configuration, implementation, retained checkpoints, household outcomes and recorded settlement artifacts. The thesis explains the economic boundary and the limits of the evaluation. These assets support a research handover. They do not remove the need for the receiving team to verify dependencies, understand the assumptions and reproduce the checks on its own environment.

The first handover requirement is a stable experiment identity. Any change to tariff bands, battery parameters, episode rules, observation features, reward or checkpoint selection defines a materially different experiment and should receive a distinct result location. Overwriting the retained run would make it difficult to distinguish an implementation correction from a research improvement. The original unsuccessful results should remain available when new variants are added. This is particularly important here because the negative learning result and the concentrated MPC gain are central findings rather than temporary outputs to discard.

The second requirement is an explicit acceptance statement for each deliverable. A dataset is accepted for this research run when it matches the declared input identity and eligibility procedure. A controller is accepted for comparison when it executes under the same physical and accounting conditions and reports its information boundary. A ledger record is accepted for the retained verification when its signature, payload, identifier and balance effect agree with the fixed stream. None of these acceptance statements means that the complete system has been certified for operation on a real electricity network.

The third requirement is separation of explanation from automation. The offline viewer presents recorded outcomes and supports a defense demonstration without rerunning training. It should be introduced as a results viewer, not as a live market or an actively optimizing battery controller. The reproducibility scripts are the route to fresh computation. This distinction matters for both a thesis examiner and a prospective collaborator: an attractive interface can make evidence easier to inspect, but it does not increase the scope of the underlying experiment.

The fourth requirement is a documented response to discrepancies. A changed file hash should trigger a version check before numerical results are compared. A mismatch between household sums and aggregate totals should be investigated at the grouping and baseline boundary. A signature failure should be treated as a failure to verify the delivered payload against the supplied key, not automatically as proof of a particular attack. A different wall-clock timing on another machine should be separated from a difference in logical balances or economic outcomes. These distinctions keep the review focused on the claim actually affected.

The fifth requirement concerns decision ownership. The energy specialist should confirm that the physical interpretation and operating assumptions are appropriate to the proposed follow-up. The finance specialist should confirm the benefit baseline, omitted cost categories and limits of extrapolation. The software specialist should demonstrate the observation timing, projection, model loading and evidence checks. The supervisor should assess whether the narrowed executed scope addresses the approved academic objectives. These are review tasks that can be assigned, not facts about completed contributions or approvals that the software can infer.

Finally, a continuation decision should specify what evidence would justify stopping or redirecting the project. If a better-covered, longer-horizon baseline still offers negligible economic opportunity after relevant costs, adding a more complex learner may have little practical value. If a fairness constraint makes participation infeasible, the coordinator must revisit the service or compensation design. If a single trusted operator satisfies the record-keeping requirement, a distributed ledger needs a separate justification. Such outcomes would be legitimate conclusions of the evaluation rather than failures to produce a sufficiently sophisticated technology demonstration.

8.10 Polish market application and transfer limits

The research design dated 21 September 2026 chooses a BESS trader in Poland as the intended beneficiary. Its objective is expected net arbitrage profit, with a proposed penalty or constraint on downside loss. The modeled asset has 1 MW of power and 2 MWh of nameplate energy. This target replaces a broad product vision, but it does not change the provenance of the measured Irish results. The current thesis therefore records a completed community demonstrator and a market application whose empirical validation is pending.

The intended market is the Polish bidding zone within Single Day-Ahead Coupling. SDAC uses EUPHEMIA to match orders subject to market and cross-zonal constraints (ENTSO-E, n.d.). The transition to quarter-hour products took effect for delivery on 1 October 2025 (PSE, 2025). These sources establish the institutional setting, not market membership, order acceptance or settlement compliance for GridMind. The implemented preparation assumes a price-taking schedule accepted in full. Partial acceptance and resulting imbalances are not simulated.

The proposed source is ENTSO-E Transparency Platform energy prices for zone 10YPL-AREA-----S. Candidate dates are October 2025–May 2026 for training, June 2026 for validation and July–August 2026 for final evaluation. These dates are not certified untouched. Raw extraction, access history, units, publication versions, exclusions and hashes must be audited before the experiment is frozen. API access requires a registered security token (ENTSO-E Transparency Platform, 2026); no Polish market dataset has been acquired or scored in the delivered package.

Historical household profiles cannot substitute for these prices. The Irish source has no verified offset for joining an external market series, uses assumed retail-like tariffs and applies four-hour inventory resets. The Polish design uses a complete local delivery day and day-ahead decisions. Likewise, the six-household fairness reward does not instantiate a trader’s CVaR objective. These differences prevent direct transfer of monetary gains, risk estimates or controller rankings.

8.11 Verified scheduling and information boundaries

A new Python module implements a deterministic mixed-integer scheduling baseline for the Polish design. It maximizes the sum of price times net export minus an AC-throughput charge. The default throughput charge is 10 EUR/MWh in each direction, a scenario assumption rather than an estimated lifetime cost. Charge and discharge

efficiencies are each 0.95; stored energy remains between 0.2 and 1.8 MWh and returns to its initial 1 MWh. Power and net connection exchange are bounded by 1 MW. The usable energy range is consequently 1.6 MWh, not the 2 MWh nameplate value.

For interval t, the balance is $E(t+1) = E(t) + 0.95\,P_{ch}(t)\,\Delta t - P_{dis}(t)\,\Delta t / 0.95$. A binary operating-mode variable prevents simultaneous charge and discharge, including during negative-price intervals. The solver uses SciPy's HiGHS MILP interface (SciPy, n.d.). This deterministic formulation has no CVaR constraint. Ten independently owned batteries and owner-level availability cannot be inferred from an aggregate feasible trajectory; physical disaggregation and owner settlement remain required for that version of the proposal.

The time interface builds a unique UTC index from the Europe/Warsaw delivery date. It admits 92, 96 or 100 quarter-hours on short, ordinary or long clock-change days. It rejects missing, duplicated, unordered or hourly rows instead of silently repeating hourly prices. Forecast issue timestamps must precede the modeled D−1 noon gate. This is a software boundary check on supplied metadata, not independent attestation of historical data availability. A later revised forecast cannot be made causal merely by assigning an earlier timestamp.

A command-line evaluator optimizes from the forecast file before opening the realized-price file. It then scores that fixed schedule and retains file hashes, battery assumptions and output trajectories. Complete acceptance, zero default ledger monetary charge and absence of imbalance/network charges are explicit evaluation assumptions. Gas counts from a local EVM are not converted into a fabricated fee. Analytical tests verify the idle solution at flat positive prices, a two-interval arbitrage solution, negative-price exclusivity, grid limits and terminal inventory. A deliberately reversed two-price forecast changes realized value from +25 to −25 EUR in a unit-efficiency test fixture; these are correctness checks, not empirical market returns.

8.12 Independent evaluation and remaining scientific decisions

A defensible final Polish study should lock one primary comparison: daily net profit of the selected forecast-conditioned PPO policy minus the deterministic MILP under the same forecast, physical limits and cost assumptions. Forecasting needs seasonal naive, regularized linear and gradient-boosting comparators before a Transformer is credited with value. Equal decision information is essential; perfect future prices belong only in

a separate oracle arm. These model names in the design are planned comparisons, not claims that they have been trained on Polish data.

The protocol must record data access before any held-out policy scores are calculated. Hyperparameters, transformations, predictor selection, reward scale, risk coefficient, training seeds and stopping rules belong to training/validation. If the July–August candidate window has already influenced choices, another period is needed. Merely assigning a test label or hashing a file after inspection does not restore independence. A locked final configuration must be evaluated once, while failed runs and exclusions remain in the archive.

Daily paired profit differences should be the statistical unit for the primary market comparison. A block resampling design and its block length should be chosen using development data, with sensitivity reported for temporal dependence. Seed variation is a different uncertainty source and must not multiply the number of test days. Sixty-two candidate test days yield only 3.1 tail-day equivalents at 95 percent CVaR. A precise risk claim therefore needs a broader regime sample or explicitly wide uncertainty. Sample size should be justified by an effect and precision target rather than a generic number of days.

Forecast value should be measured by replacing the forecast while holding the controller fixed, then by retraining otherwise matched policies with and without forecast inputs. These answer different questions. Realized price sequences must not enter a day-ahead policy through a within-day observation update. A joint study also needs chronological price versioning, exchange and imbalance assumptions, and an auditable allocation from aggregate dispatch to owner delivery. The present MPC forecast intervention provides an implemented template for the first comparison, not completion of all these requirements.

The blockchain hypothesis must be conditional on governance. With one trusted administrator, the signed database already supplies authenticated records and transactional consistency. A multi-owner claim needs independently administered participants, key ownership and failure assumptions, plus equivalent database controls. Meter signatures authenticate the reporting identity but do not establish physical truth. No current result establishes escrow, tamper-proof meters, decentralized validator resilience or a legal settlement service. Those properties must be implemented and tested separately if retained in the final claim.

9 Conclusions

9.1 Achievement of the research objectives

The completed empirical work implements local allocation, constrained batteries, causal forecast-to-MPC coupling and transaction accounting with measured Irish profiles. Ten original learned models and sixty follow-up checkpoints are retained. The new Polish scheduling and clock components pass analytical checks, while market-data validation remains pending. These evidence levels are explicitly separated. A further 501 later-period episodes evaluate all sixty frozen follow-up models against five baseline controllers without retraining.

The initial learned policies do not beat persistence MPC. At 300 updates, all ten discrete policies reproduce idle sharing; continuous policies and the tested fairness penalty do not establish a consistent advantage. Ridge MPC achieves lower total cost than persistence MPC on the reused period, despite slightly worse first-step MAE, and leaves two households worse off than idle. Thus prediction accuracy, community efficiency and individual benefit require separate evaluation.

9.2 Answers and limits

The prototype can preserve the tested energy, storage and cash identities. It can compare control policies using the same episode boundaries and reproduce accounting balances across two ledger implementations. It cannot establish general learning superiority, complete real-world grid feasibility, commercial return, legal permission to trade, or robustness of public-chain consensus. The wider title is therefore interpreted as an investigation of integration and risk diagnostics, not as evidence that autonomous wholesale trading with optimized risk has been achieved.

The original fourteen-day pilot and forty-four June battery episodes are supplemented by 501 July–December episodes across 168 retained days. The later window covers 45.38% of possible blocks. These remain dependent observations of six households. The reconstructed protocol discloses prior partial access, so the larger study improves temporal coverage without becoming a confirmatory test. No enforced daily risk constraint is available.

9.3 Final conclusion

The contribution is an auditable evaluation of community battery control and settlement. Controlled interventions expose the limits of the tested learners and the sensitivity of

forecast value to the evaluation period. Frozen models, paired temporal comparisons and household outcomes make those limits inspectable. Physical correctness, economic value, distribution and settlement trust remain distinct claims. Independent data and academic review of the stated scope are still necessary.

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Appendix A Reproduction and evidence

Appendix B Contribution record

Appendix A Reproduction and evidence

The companion archive contains source code, processed measurements, ten original and sixty follow-up checkpoints, and recorded results. Paths below are relative to gridmind/. Evaluation-only commands replay saved policies. The offline viewer shows the original experiment; follow-up evidence is in results/ablations/.

data/source_manifest.json — Source identifiers, original hashes and licensing record.

data/storenet_h1_h6_15min.csv — Processed measured input for the delivered period.

configs/extension_protocol.json — Battery, split, tariff and training assumptions.

results/pilot_results.json — Allocation and forecasting summaries.

results/extension/control_results.json — Control outcomes, checkpoint selection and residuals.

results/extension/household_episode_results.csv — Per-household results including losses.

results/extension/models/ — Ten trained IPPO/MAPPO checkpoints.

contracts/compiled.json — Compiler input, ABI and deployed bytecode.

results/extension/ledger_results.json — Recorded checks, gas and balance parity.

results/extension/signed_settlements.sqlite — Signed transaction log.

docs/EXTENSION_REPRODUCTION.md — Installation and replay commands.

scripts/verify_defense_evidence.py — Read-only hash and recorded-output consistency checks.

demo/defense_viewer.html — Offline recorded-results viewer.

configs/ablation_protocol.json — Fixed reward and action/budget comparison settings.

results/ablations/ — Sixty checkpoints, per-seed summaries and household outcomes.

docs/ABLATION_REPRODUCTION.md — Training, replay and verification commands.

The research unit-test suite contains 37 checks: the original 24, four reward and policy checks, and nine market and forecast-coupling checks. Ten recorded ledger checks cover rejection, rollback and signatures. Passing these checks does not certify the legacy API, website, physical feeder or real-world market operation.

Run python scripts/analyze_defense_results.py to reproduce results/extension/defense_analysis.json from saved records. Run python scripts/verify_saved_ledger.py to verify retained signatures and reconstruct balances;

extension dependencies are required. Neither command retrains policies or re-runs an EVM.

Forecast coupling: PYTHONPATH=src python -m gridmind_research.forecast_control. Read configs/forecast_control_protocol.json before rerunning; outputs replace only this experiment’s directory.

Follow-up audit: python scripts/verify_follow_up.py. It checks sixty checkpoints, retained replay results and the four forecast arms without new training. Read docs/ABLATION_REPRODUCTION.md and docs/POLISH_MARKET_STUDY.md for scope and input contracts.

Later-period reproduction: scripts/evaluate_later_period.py, configs/later_period_protocol.json and docs/LATER_PERIOD_REPRODUCTION.md. The archive includes full-year processed inputs, eligibility for all 1,104 candidates, execution hashes, all 65 arm outcomes and scripts/analyze_temporal_sensitivity.py.

Appendix B Contribution record

The reported team roles are Taha Can Tilkioğlu for AI, Serhat Yiğit for energy and Cem Yılmaz for finance. These intended responsibilities do not certify actual authorship of every section or computation. Each member must review and declare the work personally performed, including use of computational and writing assistance under the institution's rules.

Taha Can Tilkioğlu, student number 150224: AI and software scope. Before declaration, personally reproduce the saved-policy evaluation and explain observations, reward, checkpoint selection and code changes. Record the exact sections reviewed or authored and the evidence of review.

Serhat Yiğit: energy scope. Student number and actual contribution declaration remain to be completed. Review quantities, units, time conventions, battery parameters, the proposed Polish system and physical limits.

Cem Yılmaz: finance scope. Student number and actual contribution declaration remain to be completed. Review outside options, fees, profit and benefit definitions, tail risk and investment limitations.

AI-assisted content is present throughout all chapters, both abstracts, tables, figures, source code, computed analyses and defense materials. ChatGPT/Codex assisted drafting, implementation, execution and preparation, rather than only proofreading. Retained computation establishes reproducibility of the reported numerical outputs; it does not certify independent student authorship. This record does not certify supervisor approval. The team must document its own review and contributions under the university's GenAI rules (Poznań University of Technology, 2024).